Online Supplement to "Feeding the City: How Rural Productivity Shapes Urban Structural Change"

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This supplement reports data definitions, coverage and geographic construction, additional identification and inference checks, employment and food-price results, and urban-emergence evidence. Service-flow measurement and model reference material follow the empirical results. Table, figure and section numbers beginning with S refer to this supplement; other references identify the paper and its main appendix.

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S1 Data, Outcomes, and Coverage

S1.1 Statistical Sources and Survey Years

The empirical analysis combines three BPS statistical collections including reference years 1980, 1996, 2000, 2002, 2003, 2010, and 2011, together with the Transmigration Census replication data of Bazzi et al. (2016), which generate the exposure measure, and the geospatial travel-time and raster inputs cataloged in Table S2. The Population Census provides 2010 outcomes (population, employment by sector, migration) and 1980 baseline controls; its 2000 round supplies only the transmigration villages' agricultural employment share and settler stay share and the 2000 cohort baselines of the cohort decomposition (Appendix E.4). The Susenas household survey provides 1996 food shares, consumption quantities, calories, and unit values, 2002 consumption quantities by food item, paired with PODES 2003 production for surplus measures, and 2011 prices, wages, food shares, and housing rent, paired with the 2010 census outcome year. The PODES village survey provides 2003 production quantities by crop. Keeping consumption near the production year gives a cleaner rice surplus = production − consumption measure, while keeping prices and wages near the outcome year supports the structural-transformation interpretation. Table S1 provides a per-outcome summary including the wave / year of measurement, the analysis-level sample size (out of 187 urban centers and 2,025 subdistricts, and within the main city sample), and the construction notes. The analysis sample has 187 urban centers, every one of which has a travel time to a transmigration village (Appendix A.1), and the main city sample is the 134 centers inside the 300-minute travel-time window whose 1980 baseline the census recorded directly and whose 2010 outcomes are measured.

Table S1: Outcome Variables and Baseline Controls: Source Wave, Sample Size, and Construction. *Counts of units holding each variable, as urban centers / subdistricts, in the universe and in the main city sample.*

Outcome / variable description	**Source / wave**	**Universe**	**Main city sample**	**Construction notes**
(A) Population Census: BPS Sensus Penduduk				
Log total population and total employment, log sectoral employment and sectoral employment shares (0–1) in agriculture, manufacturing, and services, and the unemployment rate (2010)	Census 2010	186 / 2,020	134 / 405	Constructed from individual-level Sensus Penduduk 2010 records aggregated to the urban-center and subdistrict level, with sectoral employment from the 1-digit industry classification on the individual record. Shares are sectoral employment divided by total employment, and the unemployment rate is unemployed individuals divided by labor-force participants.

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Outcome / variable description	Source / wave	Universe	Main city sample	Construction notes
Birthplace decomposition (4 exhaustive categories): share of 2010 residents born in same district / different district same province / different outer-island province / Java-Bali origin	Census 2010	186 / 2,020	134 / 405	Categories are mutually exclusive and exhaustive, and they sum to 1.
1980 baseline controls: log total population, log total employment, urban-population share, log agricultural employment	Census 1980	183 / 1,897	134 / 384	Conditioning controls in the headline reduced-form specification.
(B) Susenas 2011: prices, wages, food share, housing rent (aligned with the 2010 outcome year)				
Household food share of total consumption (2011)	Susenas 2011	186 / 1,954	134 / 397	Household-weighted mean of food expenditures divided by total consumption expenditures (0–1).
Median monthly wage: all sectors, agricultural workers, non-agricultural workers (2011)	Susenas 2011	186 / 1,948	134 / 388	Constructed at the subdistrict level from the Susenas 2011 individual labor module by taking the 50th percentile of monthly wage income for currently employed workers. The wage item is net monthly earnings from the main job, in money and goods, and is asked of everyone who worked, so the earner base includes the self-employed and own-account farmers; unpaid family workers, who do not report earnings, drop out of the median.
Log nominal wage and log real (price-deflated) wage (2011)	Susenas 2011	186 / 1,951	134 / 397	Log transforms of the subdistrict-level median wage. The real (price-deflated) version is constructed for the all-sector wage only, by dividing it by a household-expenditure-weighted price index (the agricultural-goods price index weighted by the food-expenditure share, plus the housing price index weighted by the non-food share).

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Outcome / variable description	Source / wave	Universe	Main city sample	Construction notes
Composite agricultural goods price index (broad), crops-only price index, and rice-only price index	Susenas 2011	181 / 1,945	132 / 397	Built from household item-level unit values (expenditure divided by quantity), taken to the survey-weighted median over the city's households and divided by the item's median over all urban centers' households, then aggregated across the raw agricultural food items into three index variants: Laspeyres (national expenditure weights), Paasche (the city's own weights), and an equal-weight geometric mean; the crops index is the geometric mean over the rice, corn and cassava items; the rice unit value is the survey-weighted value of rice consumed over the kilograms consumed, over the local and higher-quality rice items. As unit values, these indices reflect price × quality (see Appendix D.2).
Housing price index and median monthly rent (2011)	Susenas 2011	186 / 1,954	134 / 397	Imputed monthly rent for owner-occupied dwellings combined with median actual rent for renters.
Non-food price indices: fuels and utilities (2011)	Susenas 2011	177 / 1,614	129 / 349	Cost-of-living controls, constructed parallel to the agricultural goods price index from household-level unit values.
(C) Susenas 2002: consumption quantities and values (aligned with PODES 2003 production)				
Consumption per capita per year: rice quantity (kg), value (Rp), and calories (kcal), staples and legumes (kg), and coffee & tea and spices value (Rp)	Susenas 2002	142 / 877	108 / 282	Built from the Susenas 2002 food-consumption module weekly recall and scaled to annual. Staples are rice, corn, cassava, sweet potato, taro, and other tubers. The aggregate excludes wheat and sago (not produced in measurable quantities in the PODES sample). Legumes are peanut, soybean, and mungbean, and coffee & tea and spices (pepper, nutmeg, and clove) are in monetary value because the underlying items use distinct physical units.
Total dietary calories per person per year (kcal)	Susenas 2002	144 / —	109 / —	Sum of the calorie content of every item the Susenas 2002 food-consumption module records, so it measures the whole diet. Built by pooling households within an urban center, a median of 48 households per center, and excluding household-quarters that record fewer than ten food items, which are partly-recorded modules.

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Outcome / variable description	Source / wave	Universe	Main city sample	Construction notes
Subdistrict-total annual rice consumption (kg and tons), winsorized	Susenas 2002	144 / 877	109 / 282	The sum over the subdistrict's surveyed households of annual rice kilograms times the household's survey weight, and a copy winsorized at 0.5% is stored beside it.
(D) Susenas 2011 consumption by food category (constructed from the quarterly food-module raw data)				
Staples and legumes consumption per capita per year (kg), and coffee & tea and spices consumption value per capita per year (Rp), Susenas 2011	Susenas 2011	181 / 1,954	132 / 397	Constructed analogously to the Susenas 2002 aggregates and aligned with the 2010 outcome year. Quarters Q1, Q2, and Q3 are averaged then scaled to annual; Q4 was never released by BPS, so the result represents the Q1–Q3-average annualized consumption, with no fourth quarter imputed.
(E) PODES 2003: agricultural production (aligned with the Susenas 2002 consumption year)				
Rice production per capita (kg / person) and log subdistrict-level rice production (tons)	PODES 2003	182 / 1,913	132 / 373	Total rice production divided by the population of the villages with measured rice output in the same 2003 village census, and the log of absolute subdistrict-level rice production in tons.
Rice output per planted hectare (tons / ha) and planted area (ha)	PODES 2003	176 / 1,777	129 / 325	Rice output divided by planted area over the preceding year, summed over the villages with measured rice output.
Staples planted area (ha)	PODES 2003	185 / 1,955	133 / 388	Aggregate of planted area across the staples set (rice, corn, cassava, sweet potato, taro, other tubers, sorghum).
Total planted crop area (ha)	PODES 2003	184 / 1,951	133 / 389	Sum of planted area across all 26 PODES crops.
Staples production (tons)	PODES 2003	185 / 1,945	133 / 382	Aggregate of subdistrict-level production in tons for rice, corn, cassava, sweet potato, taro, other tubers, and sorghum.
Legumes production (tons)	PODES 2003	167 / 1,722	117 / 320	Aggregate of peanut, soybean, and mungbean production in tons.
Coffee & tea production (tons)	PODES 2003	133 / 1,383	100 / 261	Aggregate of coffee and tea production in tons.
Spices production (tons)	PODES 2003	103 / 1,259	72 / 225	Aggregate of pepper, nutmeg, and clove production in tons.

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Outcome / variable description	Source / wave	Universe	Main city sample	Construction notes
(F) Derived surplus outcomes (PODES 2003 production minus Susenas 2002 / 2011 consumption)				
Rice and staples surplus (tons): subdistrict production minus subdistrict consumption	PODES 2003 – Sus 2002	143 / 833	108 / 259	The surplus subtracts, from production in milled weight, the consumption per person of the surveyed subdistricts times the 2003 population of the villages with a rice output figure. The rice-import figure uses consumption minus production and restricts both terms to surveyed subdistricts with reported rice output. The staples surplus is PODES 2003 production of rice, corn, cassava, sweet potato, taro, other tubers, and sorghum minus Susenas 2002 consumption of rice, corn, cassava, sweet potato, taro, and other tubers. The consumption aggregate does not include a separate sorghum item.
Legumes surplus (tons)	PODES 2003 – Sus 2002	130 / 759	95 / 222	Production minus consumption for legumes (peanut, soybean, mungbean).
Staples surplus using 2011 consumption (tons) [robustness]	PODES 2003 – Sus 2011	182 / 1,849	132 / 366	Robustness variant: pairs PODES 2003 production with Susenas 2011 consumption, where the main rows above use Susenas 2002.
Legumes surplus using 2011 consumption (tons) [robustness]	PODES 2003 – Sus 2011	167 / 1,666	117 / 313	Analog of the staples-2011 surplus for legumes.

Notes: Each count is the number of units holding every measure a row names, as urban centers / subdistricts. The universe is the 187 urban centers of the analysis universe and the 2,025 outer-island subdistricts in year-2000 boundaries; the main city sample is its 134 cities and the 405 subdistricts that make them up. A city enters the 1980 row when the census recorded its 1980 baseline directly, and a share or a rate counts where its numerator and a positive denominator are present. The rice-only price index is built for urban centers only, so the subdistrict counts of the price-index row cover the broad and crops-only indices; a dash marks a measure built for urban centers only.

S1.2 Earlier Household-Survey Outcomes

The 1996 Susenas wave supplies food expenditure shares, quantities, calories, and unit values. Food indices are constructed from household item-level expenditure and quantity; the rice comparison distinguishes local and higher-quality rice. The food-price, rice-quality, and price-change exhibits report their own samples and transformations. The price-change comparison uses cities observed in both 1996 and 2002. The coverage comparison in Table A5 distinguishes survey-wave availability from the number of observations in each outcome regression.

S1.3 Coverage by Source and Sample

Susenas 2011 outer-island subdistrict-level coverage is 1,951 of 2,025 subdistricts (96.3% by count) but 98.7% of the outer-island population, with a strong urban-bias by sampling design: subdistricts in the largest population quintile are 1.1 times more likely to be sampled than those in the smallest, and subdistricts with $\geq 52\%$ urban population are essentially fully covered while those with $< 15\%$ urban are only 95.4% covered. The 2011 sample is appropriate for the structural-transformation analysis but describes the urban / kotamadya subset of the subdistrict panel. The 1,951-subdistrict figure describes the prices-and-wages module; the consumption-by-food-category aggregates in block (D) of Table S1 come from the quarterly food-consumption module, whose coverage is much wider (1,954 of 2,025 subdistricts), which is why that block's sample counts differ from the coverage stated here. Susenas 2002 outer-island subdistrict-level coverage is 877 of 2,025 (43.3% by count, 61.6% by population). Aceh has zero Susenas 2002 coverage, a pre-existing structural coverage gap in the survey-frame design.

S1.4 Crop Units and Missing Observations

For the PODES 2003 crop totals, a total is set to missing when at least one village observation is absent and the observed village values sum to zero. Otherwise the observed sum is retained with a count of absent village observations; the same rule applies when subdistrict totals are summed to cities. An observed zero is retained when all component observations are present. Rice production used in the production and consumption comparison is expressed in milled weight. Susenas 2002 subdistrict-level rice consumption is winsorized at the 0.5% level for the surplus calculation. The Susenas 2011 consumption-by-food-category aggregates are not winsorized at the variable-construction stage.

Coffee & tea and spices surplus measures not constructed. PODES 2003 production for coffee, tea, and spices is recorded in tons; Susenas 2002 and 2011 consumption for the same items is recorded in monetary value (Rupiah), because the underlying items use distinct physical units across coffee, tea, pepper, nutmeg, and clove. There is no clean tonnage conversion without auxiliary price imputation, so the corresponding surplus measures are not constructed. Log production and log consumption value are reported as separate outcomes instead.

Land measure for the agricultural production function. Several subdistrict-level land measures are available from PODES 2003: rice planted area, staples planted area, and total planted crop area, plus additional sub-aggregates for legumes (peanut, soybean, mungbean), plantation crops (rubber, palm oil, coffee, tea, cocoa), coffee & tea on their own, and spices (pepper, nutmeg, clove), each summed from the individual-crop planted-area variables. Individual planted-area variables are also retained separately for each of the 26 PODES crops (the staples and legumes items listed above, plus coffee, tea, pepper, nutmeg, clove, rubber, palm oil, cocoa, sugar palm, sugarcane, cashew, cinnamon, cotton, vanilla, tobacco, jute). The land endowment that enters the model, $\bar{T}_j$, is crop area plus the satellite-derived built-up surface (GHS-BUILT-S), at the model-unit level (see the model section). The PODES harvested-area measures remain available as agricultural-land alternatives for robustness. The rice-only measure sharply understates productive land in plantation-dominant outer-island districts (especially

Sumatra and Kalimantan), the total arable measure is the broadest, and staples is an intermediate option.

S2 Geographic Units and Exposure Measurement

S2.1 Census Links Across Administrative Boundaries

Both the 2000 and the 2010 censuses record each person's village, so the link between the two vintages is made at the village level and then aggregated. I use the BPS village-history file (*Histori Master File Desa*, 1998-2014), which has one row per village with that village's administrative code in every vintage, so a 2010 census village and its year-2000 predecessor appear on the same row. Each 2010 village is counted in the year-2000 subdistrict of its year-2000 predecessor.

Three rules govern the matching. First, only rows that are a village in both vintages are used, since the file also includes subdistrict and district heading rows. Second, a 2010 village whose year-2000 predecessors lay in two different subdistricts is left unresolved; this affects five villages nationally. Third, where the file does not list a village, the village keeps the assignment from the boundary overlay, so no village is dropped.

The file covers 96.3% of the 77,480 year-2010 villages, and places 45,993 of them on the year-2000 outer-island map. Two checks support the construction. First, the number of 2010 villages each year-2000 subdistrict receives is compared with the number the 2000 census recorded there: 41.7% of outer-island subdistricts are within one village of that count and 87.1% are within a factor of 0.8 to 2.0, against 22.1% and 71.5% for a geometric assignment of the same villages. Second, on the Inner Islands, which this paper does not analyze and on which the construction was not chosen, the same comparison is 95.9% against 56.4%, so the improvement is not an artifact of tuning on the estimation sample. The total 2010 population is unchanged: a village changes subdistrict and is never created or dropped.

S2.2 Spatial Inputs and Analysis Units

The empirical analysis aggregates village- and individual-level data to two spatial units: 2000-vintage subdistricts and outer-island urban centers (UCs). The mapping between the original survey-year administrative codes and the analysis units relies on a suite of crosswalks plus globally harmonized raster products. Table S2 summarizes the spatial and crosswalk inputs. Figure S1 draws each port's route by sea to the nearer of Bangkok and Ho Chi Minh City, the sea leg of the route measure in Appendix B.2.

Table S2: Spatial Boundary Definitions, Administrative Crosswalks, and Gridded Raster Inputs

Spatial input	Source	Description / use in this paper
(A) Spatial boundary definitions		
Transmigration village sites and polygons	Bazzi et al. (2016); replication archive	The exposure sample is the complete set of 814 transmigration villages in the Transmigration Census analysis sample, settled during Indonesia's Transmigration Program (1979-1988). Each village has the program-recorded year of settlement, the count of initially placed settlers, and the agro-climatic similarity of the assigned settlers.
Urban-center (UC) delineation, outer-island Indonesia	GHS Urban Center Database 2025, GHS-UCDB R2024A V1.1 (European Commission, Joint Research Centre), fixed-boundary general-characteristics layer	Outer-island urban centers aggregated from 2000-vintage subdistricts, derived from the GHS-UCDB urban-center polygons.
Functional urban area (FUA) delineation, outer-island Indonesia	GHS-FUA R2019A (European Commission, Joint Research Centre)	Wider commuting-zone polygons around the urban centers, shown in the descriptive maps.
2000-vintage subdistrict administrative boundaries	Badan Pusat Statistik (BPS), Indonesia's central statistics agency: subdistrict boundary shapefiles	2000-vintage subdistrict polygons used as the village-aggregation target and the subdistrict-level analysis unit.
(B) Administrative-code crosswalks across census vintages		
Subdistrict 1976 / 1980 to 2000 / 2002 crosswalk ("Olken crosswalk")	B. Olken, courtesy of replication / supplementary materials	Hand-built crosswalk linking 1980 Census subdistrict codes to the 2000/2002 administrative vintage.
Village 2000 to 2010 crosswalk	BPS *Histori Master File Desa* (1998–2014) village-history file; boundary overlay of 2000 and 2010 village polygons as the fallback	The primary link between the two census rounds, and the one the city outcomes use.
Subdistrict 2000 to 2010 correspondence	Derived by aggregating the village-level link above	The 2000-vintage subdistrict totals are built by summing the villages assigned above, so the subdistrict correspondence is an output of the village lineage and includes its fallback assignments.
Host-subdistrict classification	Program-village administrative codes and year-2000 census-village and subdistrict polygons	A subdistrict hosts a transmigration village when the village records it as its administrative subdistrict or more than 0.1% of the village polygon lies inside it.

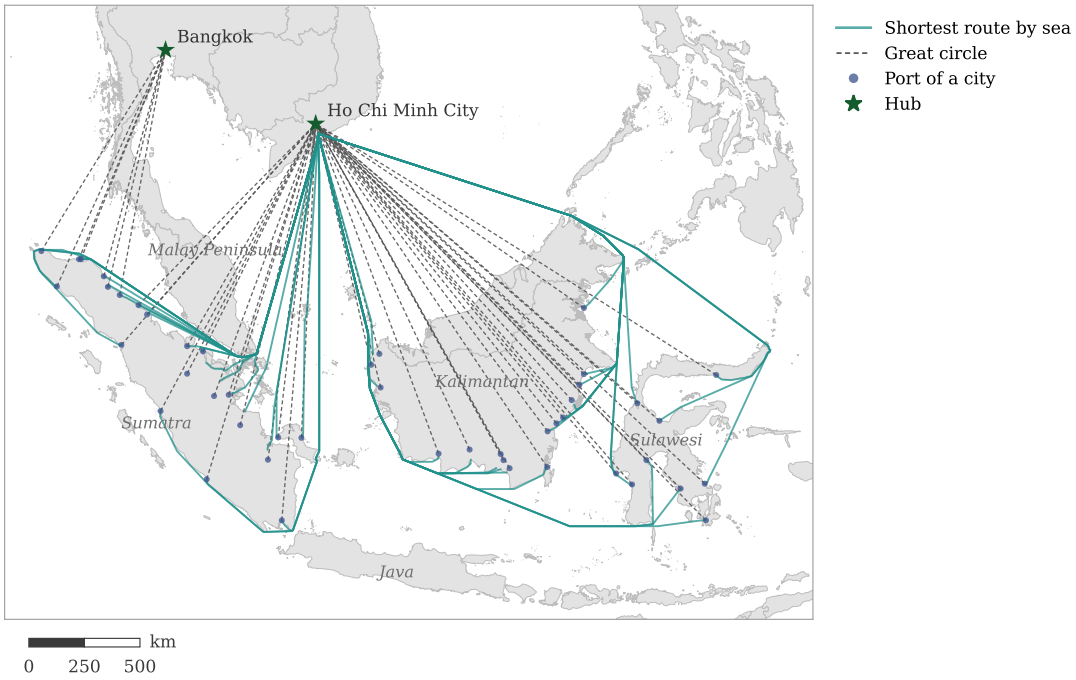
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Spatial input	Source	Description / use in this paper
(C) Travel-time and great-circle distance matrices		
Cost-distance (travel-time) matrix between UCs and transmigration villages	Constructed for this paper from an inherited general land-friction surface augmented with maritime connections; geographic alignment, pairwise calculations, and source limitations are documented in Appendix B.1	Pairwise UC×village travel-time matrix (minutes) used to construct the agro-similarity treatment as the unweighted mean over villages within the headline 300-minute window (weighted variants are robustness).
Great-circle (haversine) distance matrix between UCs and transmigration villages	Constructed for this paper from the centroid coordinates of UC and village polygons	Robustness companion to the cost-distance matrix.
(D) Gridded raster products		
GHS-POP gridded population estimates (1975, 1980, 1985, . . ., 2020)	GHS-POP R2023A (European Commission, Joint Research Centre), 100m resolution	5-year epoch population grids calibrated to the BPS Sensus Penduduk totals at the year-2000 subdistrict level.
GHS-SMOD gridded settlement-model classification (1975, 1985, 2015, 2020)	GHS-SMOD, R2022A for the 1975, 1985, and 2015 epochs and R2023A for 2020 (European Commission, Joint Research Centre), 1km resolution	Cell-level settlement-density classification (Urban Center / Urban Cluster / rural).
GHS-BUILT-S built-up-surface raster (5-year epochs, 1975-2020)	GHS-BUILT-S R2023A (European Commission, Joint Research Centre), 100m and 1000m resolution	Built-up-surface fraction per grid cell.

The census as the 1980 baseline. Every 1980 quantity in this paper comes from the population census, for three reasons. First, there is no observed satellite record for 1980. The GHSL built-up series has Landsat input epochs at 1975, 1990, 2000 and 2014, and every other epoch is interpolated between them; the 1980 population grid is close to a fixed one-third blend of the 1975 and 1990 layers, with an implied interpolation weight of 0.28 to 0.33 in every island group. The satellite 1980 baseline is interpolated between published epochs. Second, the gridded population product is not independent of the census in any case, since its counts come from census-based population estimates and the built-up layer only redistributes them within an input unit. At the subdistrict level it tracks the 1980 enumeration at a correlation of 0.57 in levels and 0.62 in logs, with the grid-to-census ratio running from 0.23 at the tenth percentile to 1.87 at the ninetieth. Third, the observed comparison epoch after the program began is the 1990 one, and it already includes the program's own settlement: within a subdistrict that split between census rounds, a successor's share of the gridded population

Figure S1: The Sea Leg to the Nearer Hub: Great Circle and Shortest Route by Sea



Notes: Each dot is the nearest port of a city of the main city sample. From each, the dashed line is the great circle to the nearer of Bangkok and Ho Chi Minh City and the solid line the shortest route by sea to the hub nearer by sea, over a 1 kilometer grid of the coastline. Stars are the two hubs. The sea leg of the route measure (Section 3.5) is the length of the solid line at 40 kilometers per hour. The scale bar is measured on the ground.

rises between 1975 and 1990 by more when that successor hosts transmigrant villages. The paper uses individual records from a 5% sample of the 1980 population census, expanded by census weights for the baseline controls.

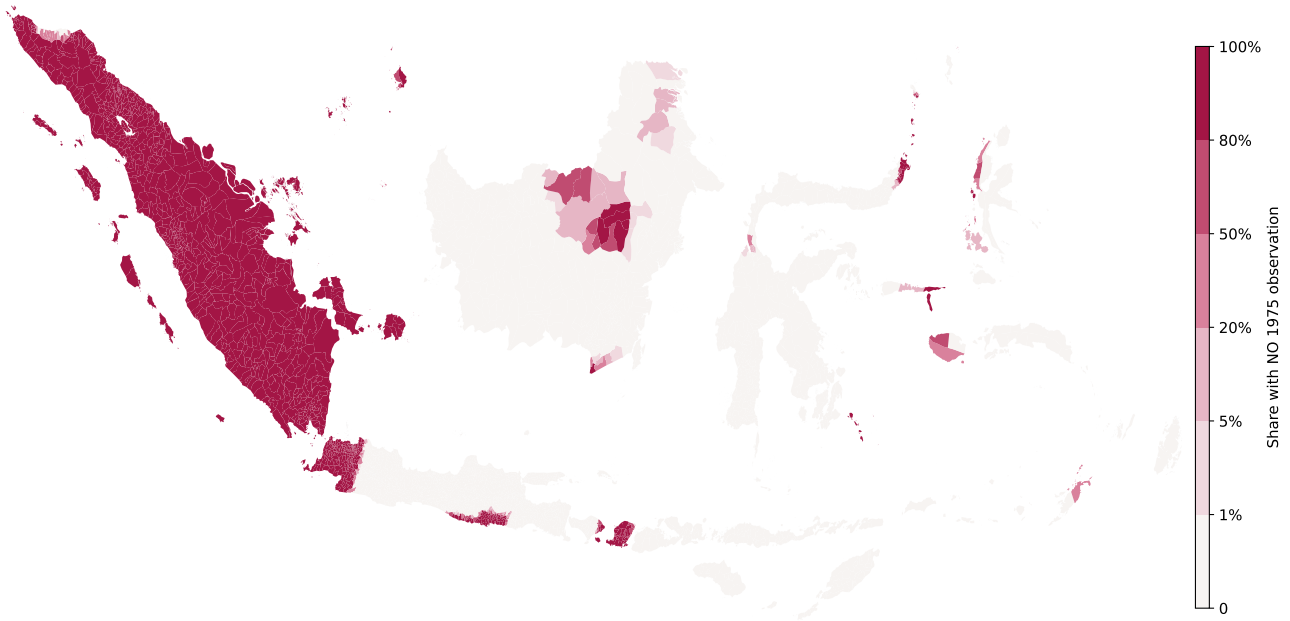
The documented sampling unit is the census block, stratified by urban and rural status with a separate stratum for institutional blocks (Central Bureau of Statistics & Minnesota Population Center 2018). Census weights expand the observed records into population and employment totals. The extract includes sample codes and block types, but the analysis does not reconstruct the full block-and-stratum sampling design or estimate city-specific sampling errors for the baseline controls. The regressions condition on the resulting baseline estimates. Alternative baseline-control sets assess sensitivity to the specification; they do not identify measurement error common to controls from the same sample. Sampling error therefore remains a limitation whose effect on the exposure coefficient is not quantified here.

S2.3 Historical Satellite Coverage

Civelli et al. (2023) report sparse 1975 Landsat coverage over much of the archipelago, Sumatra included, in an earlier GHSL release. That gap persists in the release used here, and its extent can be measured: GHSL publishes an input-count layer, `GHS-SDATA_LDS-QUANT`, giving the number of Landsat measurements behind each cell of each epoch (European Commission, Joint Research Centre 2024). On

that layer **99.3% of Sumatra's land cells have no 1975 Landsat observation at all**, against 0.5% in 1990, and 98.2% of the island's 1975 built-up surface and 97.7% of its 1975 gridded population sit on cells with none. Across the outer islands as a whole, 60.7% of 1975 gridded population is unobserved. Where imagery is absent the producers solve the built-up prediction by spatio-temporal interpolation whose dynamic component is drawn from the epochs that *were* observed, which here begin in 1990, after the program. The satellite baseline therefore includes interpolated values informed by settlement recorded after the program. The census does not have this problem. It measures 1980 directly.

Figure S2: Spatial Coverage of Landsat Observations for the 1975 Epoch



Notes: Year-2000 subdistrict boundaries, shaded by the share of the unit's land cells with no Landsat measurement behind the 1975 epoch. 889 of 921 Sumatran subdistricts are entirely unobserved and 909 of 921 have under 80% of their land cells observed, the threshold Civelli et al. (2023) use to mark a village as missing. By island group, the shares of land cells with no 1975 observation, of 1975 built-up surface on such cells and of 1975 gridded population on such cells are 99.3%, 98.2% and 97.7% in Sumatra (47.743508 million land cells), 17.7%, 26.5% and 20.4% in Java and Bali (13.919709 million), 6.5%, 28.6% and 34.8% in Nusa Tenggara (6.727677 million), 5.5%, 1.1% and 0.9% in Kalimantan (53.835817 million), 1.5%, 3.8% and 3.6% in Sulawesi (18.731243 million), and 8.1%, 16.9% and 13.9% in Maluku (7.822973 million). Over the outer islands (every group except Java and Bali), the estimation region, they are 38.4%, 53.5% and 60.7%. The counts are computed from the Global Human Settlement Layer's record of how many Landsat measurements support each 100-meter cell of its 1975 layer, summed over the land cells of year-2000 subdistricts. A cell with none has 1975 built-up and population assigned by spatio-temporal interpolation. Nationally the mean is 0.83 measurements per land cell for 1975 against 1.19 for 1990, 1.26 for 2000 and 1.43 for 2014. Source: the Global Human Settlement Layer, 2023 release (European Commission, Joint Research Centre 2024).

The gap follows the coverage of the Landsat archive itself. It is not scattered cloud cover: 61.7% of the variance in the subdistrict-level unobserved share sits between island groups and only 5.8% within a district. Sumatra, where the program placed most settlers, is the region with almost no record at all, and Indonesia had no Landsat receiving station of its own during the Multispectral Scanner era (Figure S2).

A gridded product reports a value for every cell in every published epoch, and nothing in the raster distinguishes a measurement from a model output. The grid is complete even where nothing was

measured. So a 1980 satellite baseline for Indonesia cannot support a level comparison across places, cannot support a place-specific pre-period control, and cannot be used to validate the census, because over the treated region it is not an independent record of anything. The historical satellite archive limits these baseline uses.

A narrower use survives. Contemporaneous satellite measures, from epochs that were densely observed, remain sound: the urban-center definition used throughout this paper is a 2025-vintage product, and the built-up land measure enters only at a contemporaneous epoch. For the 1980 baseline itself the alternatives are the population census, used here through its weighted 5% sample, and the village census, PODES, which is the only other source with observations in 1980 at a fine spatial unit. The paper uses those two.

Travel-time construction. Appendix B.1 describes the inherited land-friction surface, the maritime connections added for this paper, the pairwise travel calculations, and their source-date limitations. It also presents the comparison of straight-line distance and travel-time connectivity in Figure B1a.

Allocation Within Shared Subdistricts. After mapping census counts to the common subdistrict geography, I divide a subdistrict claimed by multiple cities using fixed shares of observed 2010 village employment. Villages are assigned to the claimant with the larger built-up-polygon overlap, or to the nearer claimant when both overlaps are zero. A village without matched employment does not contribute to the weights. The same weights divide the available 1980 population, urban population, employment, agricultural employment, and other additive census counts; city ratios and logs are recalculated afterward. This assumes that each early count is distributed across the claimants in proportion to observed 2010 employment. Preserving the subdistrict totals verifies the allocation arithmetic but does not validate that historical assumption. The shared-city allocation is separate from the allocation across predecessor and successor subdistricts described below.

S2.4 Allocation Across Changed Boundaries

Administrative boundaries change between census rounds, so all spatial objects are kept on the 2000-vintage subdistrict geography, which is the only vintage digitized consistently across the sources this paper uses. Census rounds on other vintages are mapped onto it by administrative code. Where a unit split between rounds, its counts are allocated across the successor units, so that recombining a split unit's own successors recovers its enumerated total exactly. The allocation divides a split unit's counts equally among its successors. That is weaker than it first appears: an equal split reproduces a subdistrict, district, urban-center or rural-subdistrict total only when the successors of every split unit inside it stay inside it, and where they fall in different reporting areas the equal division stands in for a true split this paper does not observe. Three quarters of split units have all their successors inside the same reporting aggregate, so the approximation is confined to the remaining quarter. Weighting instead by a gridded population layer would import into the 1980 baseline a quantity that is itself correlated with where settlers were placed, and those baselines enter the estimating equation as controls.

S2.5 Village-Level Links Across Survey Years

No village-level link, administrative or spatial, spans the years between 1980 and 1996, which sets how far back village-level work can reach. Three facts set the year and the unit. Four village crosswalk families exist, and they start in three different places. Cassidy and BPS's Master File Desa panel both begin at the 1998 administrative vintage; a direct 1996 map covers 88.9% of the 1996 wave across 26 provinces, Aceh, Maluku and Irian Jaya included; and a 1980 map covers West, Central and East Java, so it holds none of the outer-island sample. No village file has a vintage between 1980 and 1996, and spatial matching does not open those years either: the earliest village map is drawn on the 2000 census, and no map of the three is complete even against the village list of its own year, holding 95.2%, 99.2% and 95.5% of the SP2000, SP2010 and PODES 2021 village lists. The subdistrict is therefore the finest unit that reaches back to 1980 everywhere the paper estimates, through the Olken crosswalk at 82 to 91% per wave, which is the level at which this paper works. Table S3 measures every route on the drives against every PODES wave.

Table S3: Village-Survey Coverage in Administrative Links and Boundary Maps

	Village, crosswalk		Subdistrict, Olken		Village, spatial	
Wave	Best file and column	Covered	Best column	Covered	Best layer	Covered
1976	Not matched	–	Not matched	–	Not matched	–
1980	PODES 80 Java `ID1980`	31.3	`id1980`	90.9	Not matched	–
1983	Not matched	–	`id1983`	88.9	Not matched	–
1986	Not matched	–	`id1983`	87.8	Not matched	–
1990	Not matched	–	`id1990`	89.5	Not matched	–
1993	Not matched	–	`id1993`	88.2	Not matched	–
1996	PODES 96-03 `iddesa_1996`	88.9	`id1997`	89.8	Not matched	–
2000	PODES 96-03 `iddesa_2000`	92.3	`id1999`	82.0	2000 census	79.9
2003	BPS MFD `id2002`	95.0	`id2002`	86.2	2000 census	77.8
2005	Cassidy `ID_SE06`	94.0	Not matched	–	2010 census	80.6
2008	Cassidy `IDPODES`	98.7	Not matched	–	2010 census	94.9
2011	Cassidy `ID2011_1`	98.8	Not matched	–	2010 census	96.3
2014	Cassidy `IDPODES14`	99.9	Not matched	–	2020 BPS	96.9
2018	Cassidy `ID2015_1`	96.3	Not matched	–	2020 BPS	96.3

	Village, crosswalk		Subdistrict, Olken		Village, spatial	
Wave	Best file and column	Covered	Best column	Covered	Best layer	Covered
2021	Cassidy `ID2015_1`	95.6	Not matched	–	2020 BPS	95.5
2024	Cassidy `ID2015_1`	90.7	Not matched	–	2020 BPS	91.5

Notes: Each row is a wave of the village census (PODES), and each "Covered" column is the share, in %, of the distinct codes of the wave found in the best-matching column of the village crosswalk named, of the subdistrict crosswalk of Olken, or of the best-matching village boundary map. The village pair reports whichever of four crosswalk families reaches furthest into the wave: Cassidy's crosswalks and the village-history panel of the national statistics agency (Master File Desa), both from the 1998 vintage; a direct 1996 to 2000 to 2003 village map over 26 provinces, the only route into the 1996 wave; and a 1980 to 2000 village map of West, Central and East Java only, which reaches 90.1% of the 1980 wave inside Java and none of the outer islands. "Not matched" means that no file of the route spans the wave (no village crosswalk has a vintage between 1980 and 1996, the Olken crosswalk ends at the 2002 vintage, and the earliest village boundary map is drawn on the 2000 census) or, in the 1976 subdistrict cell, that under 10% of the codes match, because the 1976 codes have no province part; matches outside the span of a file are suppressed. The "Village, spatial" pair matches on the code against three village boundary maps: the 2000 and 2010 census maps (66,280 and 77,480 distinct codes) and the 2020 boundary file of the national statistics agency (81,912), which hold 95.2, 99.2 and 95.5% of the village lists of the 2000 census, the 2010 census and the 2021 village census. Shading of the coverage cells steps up at 80, 90 and 95%, and cells below 80% share the shade of "Not matched". *Source:* the village census waves, the crosswalks and the boundary maps, each measured from the file itself. In the cells, BPS MFD is the Master File Desa panel of the national statistics agency (BPS), PODES 96-03 the 1996 to 2000 to 2003 village map, PODES 80 Java the 1980 to 2000 Java map, 2020 BPS the 2020 boundary file of the agency, and the entry in typewriter type is the column of the crosswalk that holds the codes of the wave.

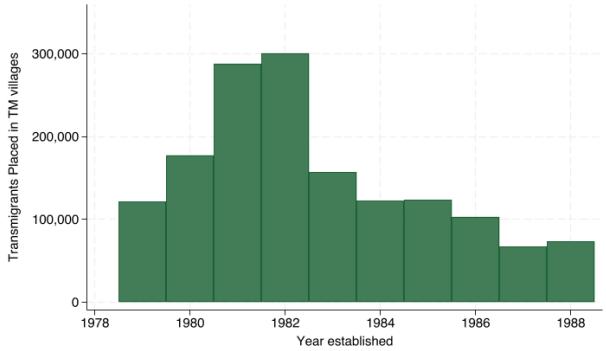
S2.6 Individual-Level Linkage Across Census Rounds

The 2000 to 2010 crosswalks above are administrative-boundary mappings and support unit-level aggregation only. Person- or household-level linkage of the 2000 and 2010 census microdata is not feasible: the 2000 census records do not contain digitized respondent names, and a full-sample probabilistic matching experiment linked only 0.08–0.43% of households at a placebo-calibrated 1% false-discovery rate. No result in this paper therefore relies on following individuals or households across census rounds; all 2000 to 2010 comparisons are aggregates within crosswalked spatial units.

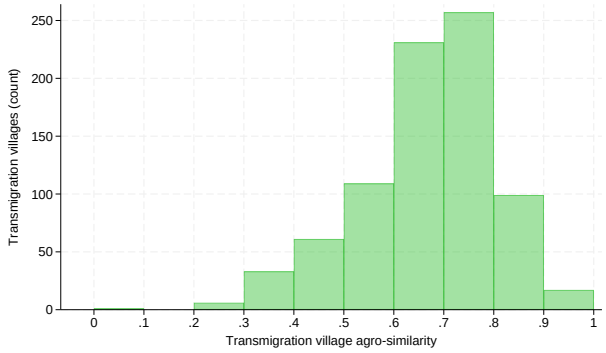
S2.7 Settlement Dates and Village Agro-Climatic Similarity

Figure S3: Transmigration Villages: Settlement by Year and Agro-Climatic Similarity

(a) Transmigrants Placed, by Settlement Year



(b) Village-Level Agro-Similarity (Transmigration Villages)



Notes: Panel (a) counts the transmigrants placed in each settlement year from 1979 to 1988 across the 814 transmigration villages. Panel (b) counts the same villages by their agro-climatic similarity to their settlers' places of origin, on its 0 to 1 scale. Source: Transmigration Census settlement records from the Bazzi et al. (2016) replication data.

S3 Additional Identification and Inference

S3.1 Common City Specification

The main city sample consists of the 134 outer-island urban centers within 300 travel minutes of a transmigration village, with a directly recorded 1980 census baseline and measured 2010 outcomes. Agro-climatic similarity is the unweighted mean across transmigration villages in that window. The city regressions control for the inverse-travel-time-weighted settler count, in thousands, and 16 destination characteristics averaged across those villages: three slope measures, ruggedness, log elevation, soil organic carbon, exchangeable sodium percentage, soil pH, two soil-texture and two drainage measures, distances to coasts and rivers, and mean rainfall and temperature over 1948-1978. The four city baselines are log population, log employment, the urban population share, and log agricultural employment in 1980. Tables state the transformations, standardization, and uncertainty method. The main city estimates use Conley standard errors allowing spatial dependence within 200 kilometers; village, lattice, and district exercises define their own samples and controls.

S3.2 Whole-Match and Origin-Composition Reassignments

The recentering of Appendix C.1 uses two permutations of the settler match. The whole-match reshuffle permutes each village's agro-climatic similarity across sites within province and Repelita settlement wave; the origin redraw holds each site's destination agro-climate fixed and redraws only the settler-origin composition.

Realized and recentered exposure are both correlated with the measured village characteristics. On the main specification's agro-climatic control set, the 16 village covariates averaged unweighted inside the same 300-minute window the exposure itself uses, the realized exposure has an R^2 of 0.775 on the estimation sample and the province-by-cohort recentered measure 0.556. Panel A of Table S4

Table S4: Agro-Similarity Recentered Against Counterfactual Settler Assignments and Its First Stages

	(1) ln pop	(2) ln emp	(3) unemp rate	(4) ln L agric	(5) agric %	(6) ln L modern	(7) modern %
Panel A: The 2010 outcomes on recentered agro-similarity							
Agro-similarity recentered $\tilde{z}^{PW}$	0.010	0.011	-0.008*	-0.037	-0.016	0.038	0.021**
	(0.050)	(0.049)	(0.005)	(0.075)	(0.011)	(0.038)	(0.010)
Agro-similarity origin-recentered $\tilde{z}^{orig}$	-0.063	-0.060	-0.005	-0.100*	-0.011	-0.006	0.027***
	(0.047)	(0.045)	(0.004)	(0.058)	(0.013)	(0.056)	(0.009)
Observations	134	134	134	134	134	134	134
Settler count	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agric Empl 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓

Panel B: First stages of standardized headline agro-similarity on the recentered exposure	Whole-match reshuffle	Origin redraw
Recentered exposure	+0.518***	+0.397***
	(0.067)	(0.067)
Bartlett spatial SE	(0.056)	(0.069)
Heteroskedasticity-robust SE	(0.044)	(0.071)
Partial R^2	0.525	0.290
Conditional SD, headline exposure	0.463	0.463
Conditional SD, recentered exposure	0.647	0.628
Cities	134	134

Notes: In Panel A, each cell is the coefficient from one regression of a 2010 population census outcome on recentered agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km, and the two rows are separate regressions. Agro-similarity recentered, written $\tilde{z}^{PW}$, is agro-similarity minus its expected value over 2,000 random reassignments of village agro-climatic similarity among the villages settled in the same province and the same five-year plan (1979 to 1983 or 1984 to 1988), standardized (the μ^{PW} of the appendix's recentering equation). Agro-similarity origin-recentered, written $\tilde{z}^{orig}$, is agro-similarity minus its mean over 2,000 draws that permute the settlers' origin composition across villages within province and settlement wave, holding each village's similarity to every origin region fixed, standardized. Outcomes are log population, log employment, the unemployment rate, log agricultural employment, the agricultural employment share, log modern-sector (manufacturing and services) employment and the modern-sector employment share. Shares and the unemployment rate are fractions. Randomization-inference p-values from the same reassignments are in columns (2) and (4) of Table S5. In Panel B, the dependent variable is standardized headline agro-climatic similarity. Each column regresses it on one standardized recentered exposure, the inverse-travel-time-weighted settler count, four 1980 baseline controls, and the 16 agro-climatic means within the 300-minute catchment, on the main city sample. Standardization follows the common-unit recentering figure and precedes restriction to that sample. The standard errors directly below coefficients use uniform spatial weights within 200 km. The next rows use Bartlett spatial weights within 200 km and heteroskedasticity-robust HC1 standard errors. Partial R^2 is the squared correlation of headline and recentered exposure after removing the controls and a constant. Conditional SD is the standard deviation of each exposure after removing the controls and a constant, in its standardized units. Check marks show the controls of Panel A. The settler count weights the settlers of each transmigration village within 300 travel minutes by the inverse of its travel time, in thousands, and the Baseline Pop 1980, Total Empl 1980 and Agric Empl 1980 rows are logs of 1980 population, employment and agricultural employment. Standard errors are in parentheses. Sources: the program-village match and the origin-destination similarity grid of Bazzi et al. (2016), using the saved 2,000 within-province-and-wave reassignments. In Panel B, stars use the normal approximation and the uniform spatial standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

subtracts the expected exposure from the treatment, reporting $\tilde{z} = z - \mu$, with coefficients expressed per standard deviation of the recentered measure. Their magnitudes are not directly comparable to the headline coefficients. The agricultural share reads -0.016 with a standard error of 0.011 and log modern-sector employment $+0.038$ with a standard error of 0.038, so of the three compositional outcomes only the modern employment share is statistically significant.

The similarity index of Bazzi et al. (2016) is linear in origin composition, $AS_v = \sum_o \omega_{ov}\, s(o, v)$, where $s(o, v)$ is origin-to-destination dyadic similarity and ω_{ov} is the share of village v's settlers born in origin district o. Their replication data provide the full dyadic grid (125 Java and Bali origins by 814 villages) and the realized migrant flows. I hold each site's similarity column $s(\cdot, v)$ fixed and assign another village's origin composition $\omega_{\cdot v'}$, permuted within province-by-wave cells, following their reassignment of origins. Transmigration villages are linked to the grid by rank in the census-code ordering. The composition-weighted grid reproduces the published village index up to an affine rescaling ($r = 1.00000$, slope 0.56), so realized and expected exposure are differenced on the reconstructed scale before standardization.

Per standard deviation of origin-recentered exposure, the modern employment-share coefficient is $+0.027$, statistically significant, and the log agricultural-employment coefficient is -0.100, marginally significant (Panel A of Table S4). The agricultural-share coefficient is -0.011 and imprecise. Log population and log modern-sector employment have negative coefficients of -0.063 and -0.006 that are not statistically significant. The assignment test using the same origin reassignments gives $p = 0.544$ for the agricultural share (Table S5). These regressions use variation in settler origins conditional on the sites. The headline also uses variation associated with site land after the linear controls; Appendix C.1 reports the components separately and discusses the qualifications from the productivity and 1980 tests.

Realized and expected exposure correlate at 0.96 under origin reassignment. Subtracting expected exposure changes both the variation and the ordering of cities. Since $\tilde{z} = z - \mu$, the correlation between raw and recentered exposure is $[\mathrm{SD}(z) - \mathrm{corr}(z, \mu)\mathrm{SD}(\mu)]/\mathrm{SD}(\tilde{z})$. Under whole-match reassignment, $\mathrm{corr}(z, \mu)$ is 0.75 and the raw and recentered measures correlate at 0.60. Under origin reassignment, the latter correlation is 0.28, with a rank correlation of 0.09; 49% of cities move from above-average to below-average exposure or the reverse, and the average city shifts 41 places among 134 cities.

Before regression controls, expected exposure under origin reassignment accounts for roughly 91% of realized exposure's variance. The agro-climatic controls explain much of realized exposure, with $R^2 = 0.775$; this regression does not separately measure how much of the expected component they explain. Province-only stratification leaves a similar variance share after recentering, 10% against 9% under province-by-wave stratification. Whole-match reassignment leaves 43% of the variance. One standard deviation of the origin-recentered exposure is about 30% of a standard deviation of realized exposure, compared with 66% for the whole-match version. Because each regression standardizes its own exposure, the coefficients describe different amounts of variation. The productivity and 1980 birthplace tests in Appendix C.1 further qualify the interpretation of the origin component.

Panel B of Table S4 reports the first stages used to put the recentering figure in headline-exposure units, with the estimator defined in the paper's appendix. Both regressions condition on the full main control set and use the same cities as that figure. Conditional standard deviations and partial R^2 describe the variation remaining after these controls. The reported standard errors allow comparison of uniform spatial, Bartlett spatial, and heteroskedasticity-robust inference. They do not test the validity of the origin or whole-match reassignment schemes.

S3.3 Assignment-Based Inference

Table S5 reports assignment-based inference for both schemes.

Table S5: Randomization-Inference p-Values by Statistic and Permutation Scheme

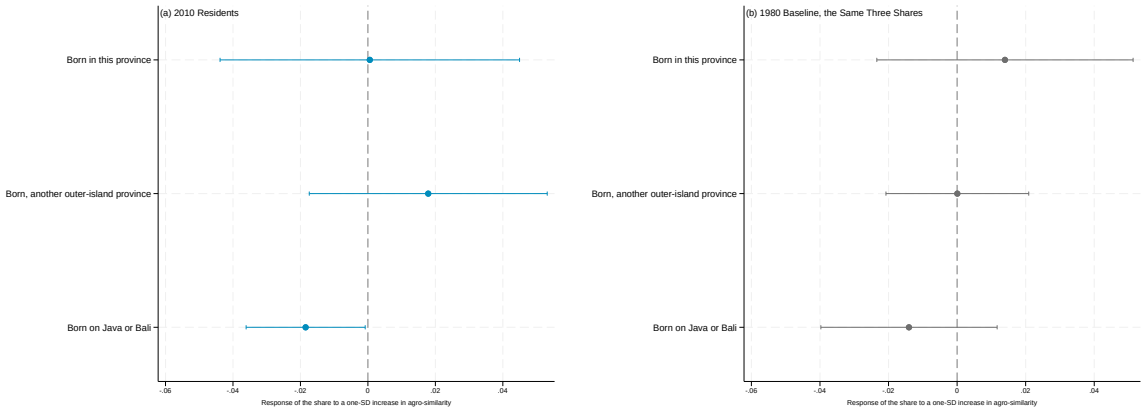
		Whole-match		Origin		Controlling for expected exposure			
	Conley	recentered		recentered		t-statistic		coefficient	
Outcome	200 km	prov. ×wave	province	prov. ×wave	province	prov. ×wave	province	prov. ×wave	province
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log population	0.389	0.827	0.883	0.210	0.534	0.666	0.670	0.833	0.837
Log employment	0.674	0.826	0.909	0.304	0.332	0.781	0.764	0.886	0.889
Unemployment rate	0.216	0.095	0.736	0.431	0.647	0.487	0.514	0.738	0.722
Log agricultural employment	0.075	0.631	0.393	0.158	0.243	0.447	0.428	0.675	0.633
Agricultural employment share	0.006	0.336	0.245	0.544	0.580	0.196	0.190	0.485	0.469
Log modern employment	0.011	0.599	0.687	0.937	0.906	0.181	0.193	0.615	0.610
Modern employment share	0.000	0.180	0.134	0.090	0.221	0.050	0.053	0.336	0.317
Observations	134	134	134	134	134	134	134	134	134

Notes: Every entry is a two-sided p-value for the null of no effect of agro-similarity on the outcome, on the main city sample. Each cell is shaded by the strength of the evidence, deepening green from the 10% to the 1% level, with pale pink for a p at or above 0.10. Column 1 is the analytic p-value from Conley SE within 200 km, behind the stars of Table 4. Columns 2 to 9 report simulation-based randomization inference over 2,000 counterfactual assignments of settlers to sites: the observed statistic is compared with the distribution of the same statistic when the settler match is re-drawn, stratified by province and Repelita settlement wave (even columns) or by province (odd columns). Columns 2 and 3 test the coefficient on the recentered exposure under the whole-match reshuffle, and columns 4 and 5 under the origin re-draw, which holds each site's destination agro-climate fixed and re-draws only the settler-origin composition. Columns 6 to 9 test the coefficient that controls for expected exposure (Table A1): columns 6 and 7 its t-statistic with a heteroskedasticity-robust (HC1) standard error, columns 8 and 9 the coefficient itself. The t-statistic is the test the appendix reports. With the expected exposures and the controls partialled out, the realized exposure has a residual standard deviation of 0.41 on the scale of the standardized treatment, against 0.20 on average across the reshuffles, and none of the 2,000 reshuffles reaches it under either stratification, so a placebo coefficient is estimated on about half the variation behind the observed one, a difference the t-statistic divides out. At the 10% level, in both columns of a pair, the whole-match recentering rejects for no outcome; the origin recentering rejects for no outcome; the t-statistic test of the controlled coefficient rejects for the modern employment share; the test of the coefficient itself rejects for no outcome. Of the cells whose pair does not reject, 2 clear 10%: the unemployment rate under the whole-match recentering stratified by province and wave (0.095); the modern employment share under the origin recentering stratified by province and wave (0.090).

S3.4 Birthplace Composition and the Baseline Census

I compare the two decompositions side by side at the only resolution both censuses support. SP1980 records place of birth at the province level, so the same-district and same-province cells of the 2010 decomposition cannot be separated at baseline; collapsing them into one same-province share makes the two panels the same object, and it is why the comparison has three categories where Figure 13 has four. The 2010 own-province coefficient is close to zero, and the other-outer-island-province coefficient is positive but imprecise. The Java-or-Bali coefficient is negative in both years; all three 1980 coefficients are imprecise (Figure S4).

Figure S4: Where Residents Were Born, 2010 Against the 1980 Baseline, at the Three-Way Resolution



Notes: Each panel regresses three exhaustive birthplace shares on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline; 134 cities in both panels), in the main specification with the full set of controls and Conley SE within 200 km. Markers are the coefficients, in share units, and bars their 90% confidence intervals from the same standard errors; the three coefficients in each panel sum to zero by construction, which both panels satisfy exactly. Panel (a) covers 2010 census residents born 1955 to 1990, with the same-district and same-province categories of Figure 13 added together; panel (b) covers all 1980 census residents at the same three categories. The two panels share one horizontal scale. Table C1 adds the 1980 shares of the same birthplace categories to the main specification.

S3.5 Exposure Restricted to Post-Baseline Settlements

Restricting the exposure to the villages settled after the count leaves it nearly unchanged and lowers the settler-count control by roughly a fifth. No city loses its hinterland (Table S6).

Table S6: Exposure From All Villages and From Post-1980 Villages

	All 814 villages	687 post-1980 villages
Exposure (unweighted mean similarity, 300 minutes)		
Mean	0.6800	0.6828
Standard deviation	0.1025	0.1016
Minimum	0.3777	0.3777
Maximum	0.9563	0.9563
Correlation with the other, levels	0.9901	
Correlation with the other, ranks	0.9762	
Villages within 300 travel minutes		
Mean per city	30.4	25.8
Median per city	12	11
Cities losing exposure entirely	0	
Settler count (inverse travel time weights, 300 minutes, thousands)		
Mean	0.466	0.365
Correlation with the other	0.8703	
Urban centers (main city sample)	134	

Notes: Computed over the main city sample. The exposure is the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, and the settler count sums their settlers weighted by inverse travel time, in thousands. The left column builds both from all 814 villages, as the main specification does; the right column from the 687 villages settled after the 1980 census, dropping the 127 placed by 1980, which hold 19.5% of all settlers. No city loses its hinterland under the restriction.

S3.6 Straight-Line Catchments and the City Footprint

When each city's catchment is redrawn as a circle whose radius is the distance to the farthest transmigration village inside its 300-minute window (a median of 157 km), log modern-sector employment rises by 0.178 log units (standard error 0.090) per standard deviation of exposure and the modern employment share by 0.040 (0.016), both statistically significant, the first slightly larger than the main estimate and the second 60% of it. The agricultural-share estimate is −0.025 (0.021), 49% of the main estimate and statistically insignificant.

The urban-center boundary the paper defines does not respond detectably to exposure. Per standard deviation of exposure, log urban-center area changes by 0.067 log units (standard error 0.065) and the log land area of a city's member subdistricts by −0.191 (0.184), both statistically insignificant, while the number of member subdistricts rises by 0.990 (0.242) on a mean of 2.8, a statistically significant estimate. More exposed cities therefore span more and smaller year-2000 subdistricts. The 1980 baselines and the 2010 outcomes are summed over the same members, so the comparison holds geography fixed across the two dates.

S3.7 Settler-Retention Proxy Definitions

The settler stay-share featured in the text divides a village's 2000-census residents born in Java or Bali by the individuals the program placed in the village, under two restrictions that each remove one class of non-settler the census can identify: residents whose recorded residence five years before enumeration was still Java or Bali, and residents born in Java or Bali after their own village was settled. The broader ratio without the two restrictions, which Figure 14 plots, counts anyone who reached the village from

the main islands on their own after the program ended, so it proxies for retention. The restrictions remove a median of 10% of the numerator, yet the conditional gradient on agro-climatic similarity moves only from 0.078 to 0.080, the definitions correlate above 0.98, and every version stays indistinguishable from zero under both the wild cluster bootstrap and Conley standard errors (Table S7). The definitions give similar estimates in this sample. No version of the numerator recovers the settlers' locally born children, who are outer-island-born and never enter it, so the measure cannot establish the retention of all original settlers and their descendants.

Table S7: Settler-Retention Proxy Under Alternative Numerator Definitions

Numerator definition	Gradient on agro-similarity	Island-clustered std. error	Wild bootstrap p-value	Conley 200 km p-value	Corr. with the plotted measure
All Java/Bali-born residents (plotted measure)	0.078	(0.106)	0.69	0.57	1.000
Excluding arrivals in the previous five years	0.080	(0.093)	0.67	0.52	0.989
Excluding those born after settlement	0.071	(0.091)	0.67	0.54	0.995
Excluding both groups	0.080	(0.086)	0.66	0.49	0.985

Notes: Each row is the slope of one stay-share definition on the agro-climatic similarity of the village itself (an index from 0 to 1), across the same 808 transmigration villages, with the seven controls of Figure 14: the shares of land with slopes of 0-2%, 2-8% and 8-30%, ruggedness, log altitude, distance to the coast, and mean 1948-1978 temperature. The stay share is the residents born in Java or Bali in the 2000 census over the individuals the program placed in the village; rows 2 to 4 remove from that numerator the residents who arrived in the five years before the census, those born after the village was settled, or both. Parentheses hold standard errors clustered by island (four clusters); the wild cluster bootstrap p-value uses Webb weights over the four islands, and the p-value uses Conley SE within 200 km over village centroids. Corr. with the plotted measure is the correlation, across the same villages, between the row's stay-share and the one plotted in the text (row 1).

S3.8 Settler-Count Controls in City Regressions

The headline main specification controls for the inverse-distance-weighted count of nearby transmigrant settlers ($SettlerCount_c^{IDW}$, in thousands) in addition to the agro-climatic similarity exposure. Because the agro-similarity measure and the migrant-count control are built over the same villages and the same travel-time window, the two are correlated, and the choice of functional form for the migrant-count control could in principle absorb some of the agro-similarity variation. This appendix reports the headline UC-level regressions under three alternative functional forms for the migrant-count control:

(i) **Linear count** (thousands of settlers, IDW): the form used in the main specification.

(ii) **Log count** (log of the IDW-weighted settler count): tests sensitivity to a constant-elasticity functional form.

(iii) **Count + count-squared** (count and its square, both reported): probes non-linearity in the count effect, motivated by the gap in significance between (i) and (ii).

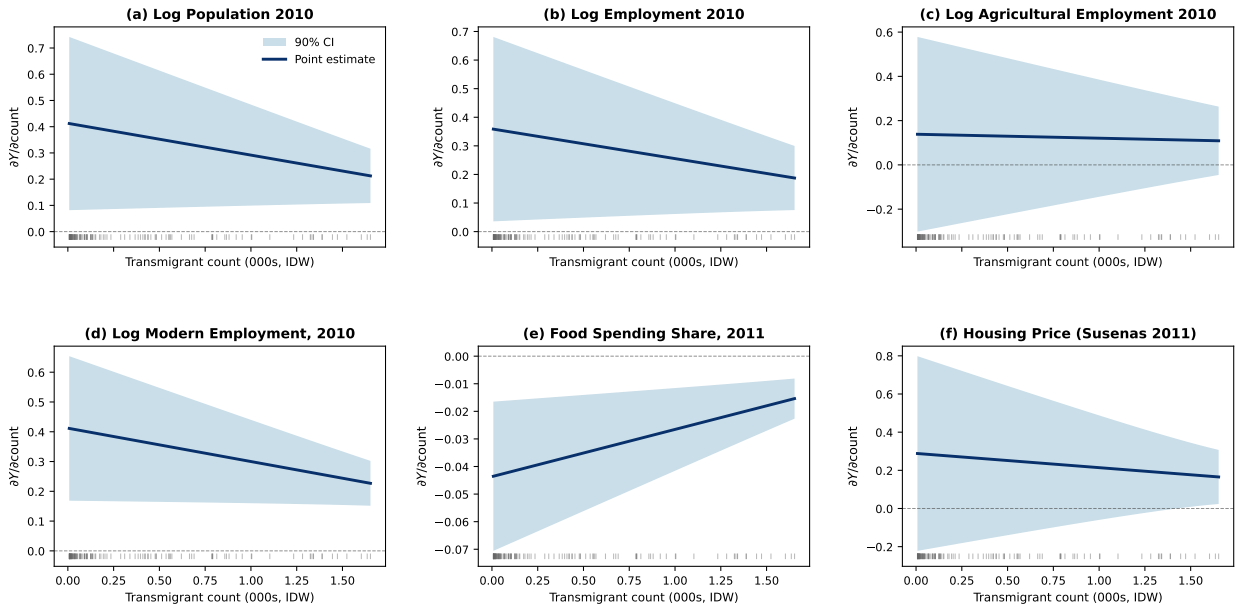
Findings. The agro-similarity coefficients are essentially invariant to the form of the settler count (Table S8 reports the linear count in Panel A, the count and its square in Panel B and the log count in Panel C). Manufacturing employment is +0.249 under the linear count, +0.266 under the log and +0.274

under count and count squared, and agricultural employment is negative and marginally significant under all three (-0.166, -0.171 and -0.164). On the seven levels outcomes the settler term itself is positive under the linear count and under the log, and significant at the 5% level on 5 of them under the linear count and 5 under the log (population, total employment, manufacturing, construction and services in both), with agricultural employment and the unemployment rate positive but imprecise: more settlers placed near a city go with a larger city and more workers in every modern sector, and the count control absorbs that. Under count plus count squared the agricultural terms are $+0.138$ and -0.009, both insignificant, so these estimates do not establish a non-monotone agricultural association; the concave shape is in manufacturing, where the count enters at $+0.604$ and its square at -0.126, both significant, and more weakly in services, whose squared term is -0.055 and marginally significant. The agro-similarity coefficients are similar across the count-control specifications examined.

The marginal effect of the settler count, $\partial Y/\partial \text{count} = \alpha_1 + 2\alpha_2 \cdot c$ from specification (iii), traces how the response changes across the empirical range of settler counts, with 90% Conley confidence bands, for population, total employment, agricultural employment, modern-sector employment, the household food-expenditure share, and housing price. For population, total employment and modern-sector employment the marginal effect is positive across the whole plotted range, the 5th to the 95th percentile of the settler count, and largest where settlement is thinnest: the estimated marginal association decreases as the settler count rises, and the curves flatten without turning down inside the range. Agricultural employment does not respond detectably at any count. The household food-expenditure share falls with the count, by less as settlement thickens. In the count and count-squared estimates for the levels outcomes (Table S8, below) the squared term is negative on every employment outcome, significant for manufacturing and marginally significant for services. Figure S5 plots these marginal effects with their bands.

Count and count-squared estimates for the main outcomes. With the settler count and its square as controls, the agro-similarity coefficients on the agricultural employment share and on the modern-sector employment share keep their sign and magnitude and remain significant. Panel B of Table S8 reports this specification for the seven outcomes of Table 4, estimated on the same cities.

Figure S5: Conditional Marginal Associations With the Settler Count



Notes: Each panel plots the marginal effect of the settler count, $\partial Y/\partial \text{count} = \alpha_1 + 2\alpha_2 c$, from one regression on the main city sample of the outcome on agro-similarity, the settler count within 300 minutes (thousands, weighted by inverse travel time) and its square, 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. For population, employment, and agricultural employment the two count coefficients are those of Table S8. The effect is evaluated at 30 points from the 5th to the 95th percentile of the settler count across the 138 urban centers with a transmigration village within 300 travel minutes, tick marks along the bottom show their counts in that range, and the shading is the 90% confidence interval from Conley SE within 200 km. Outcomes are 2010 census log population, log employment, log agricultural employment and log modern (manufacturing plus services) employment, and the 2011 household-survey (Susenas) mean household food expenditure share and housing price index, the weighted median rent of the urban center relative to the national level.

Table S8: Population and Sector Employment in 2010 Under Three Forms of the Settler Count

	Population + Employment			Agric Empl		Log sector employment			Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	ln pop	ln emp	unemp rate	ln L agric	agric %	L manuf	L constr	L serv	ln L modern	modern %
Panel A: The settler count										
Agro-similarity	0.051	0.024	0.008	-0.166*	-0.051***	0.249**	0.050	0.144**	0.158**	0.066***
	(0.059)	(0.057)	(0.007)	(0.093)	(0.018)	(0.120)	(0.067)	(0.067)	(0.062)	(0.015)
Settler count (inverse	0.215***	0.189***	0.005	0.109	-0.022	0.191**	0.292***	0.241***	0.229***	0.009
travel time weighted)	(0.063)	(0.068)	(0.005)	(0.095)	(0.016)	(0.096)	(0.052)	(0.049)	(0.053)	(0.013)
Panel B: The settler count and its square										
Agro-similarity	0.063	0.034	0.009	-0.164*	-0.052***	0.274**	0.059	0.155**	0.169***	0.066***
	(0.060)	(0.058)	(0.006)	(0.093)	(0.018)	(0.120)	(0.072)	(0.068)	(0.063)	(0.015)
Settler count (inverse	0.413**	0.359*	0.021	0.138	-0.042	0.604**	0.438***	0.420***	0.412***	0.006
travel time weighted)	(0.201)	(0.197)	(0.017)	(0.268)	(0.039)	(0.251)	(0.163)	(0.145)	(0.148)	(0.037)
Settler count squared	-0.061	-0.052	-0.005	-0.009	0.006	-0.126**	-0.045	-0.055*	-0.056*	0.001
	(0.043)	(0.040)	(0.005)	(0.055)	(0.009)	(0.056)	(0.035)	(0.032)	(0.032)	(0.008)
Panel C: The log settler count										
Agro-similarity	0.052	0.028	0.008	-0.171*	-0.053***	0.266**	0.041	0.150***	0.164***	0.068***
	(0.052)	(0.050)	(0.007)	(0.096)	(0.018)	(0.125)	(0.064)	(0.055)	(0.053)	(0.015)
Log settler count	0.089***	0.087**	0.001	0.029	-0.014**	0.129***	0.090***	0.114***	0.109***	0.008
	(0.034)	(0.034)	(0.002)	(0.046)	(0.006)	(0.038)	(0.031)	(0.032)	(0.031)	(0.006)
Outcome mean	11.842	10.933	0.122	9.822	0.374	7.556	8.011	10.136	10.232	0.516
Outcome SD	0.854	0.840	0.040	0.896	0.183	1.356	1.111	0.966	0.975	0.156
Observations	134	134	134	134	134	134	134	134	134	134
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Agric Empl 1980	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Each column is a separate urban-center regression on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Panel A enters the settler count within 300 minutes, weighted by inverse travel time and in thousands, as in the main specification. Panel B adds its square. Panel C enters the log of one plus that count in settlers. Outcomes are from the 2010 population census (the unemployment rate and the two employment shares are fractions), and modern-sector employment is manufacturing plus services. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S3.9 Settler-Count Controls in Rural Public Provision

Leaving the settler control (for a subdistrict with a settlement, the number of settlers placed in the transmigration villages that overlap it, and for one without, the settlers within 300 travel minutes, weighted by inverse travel time) out of the regressions of public provision in rural subdistricts with and without a settlement, or counting the settlers within 300 travel minutes for the subdistricts with a settlement too, changes the stars of 2 and the dagger of 1 of the 56 corresponding estimates on the sensitivity sample, and the largest move of an estimate is 0.16 of the main specification's Conley standard error. The figures leave West Kalimantan out of their 1980 to 2011 presence-share rows, and this check keeps it (Figure S6 and Figure S7).

Figure S6: Public Provision and Roads in the City and Its Countryside

(a) By Area of the City

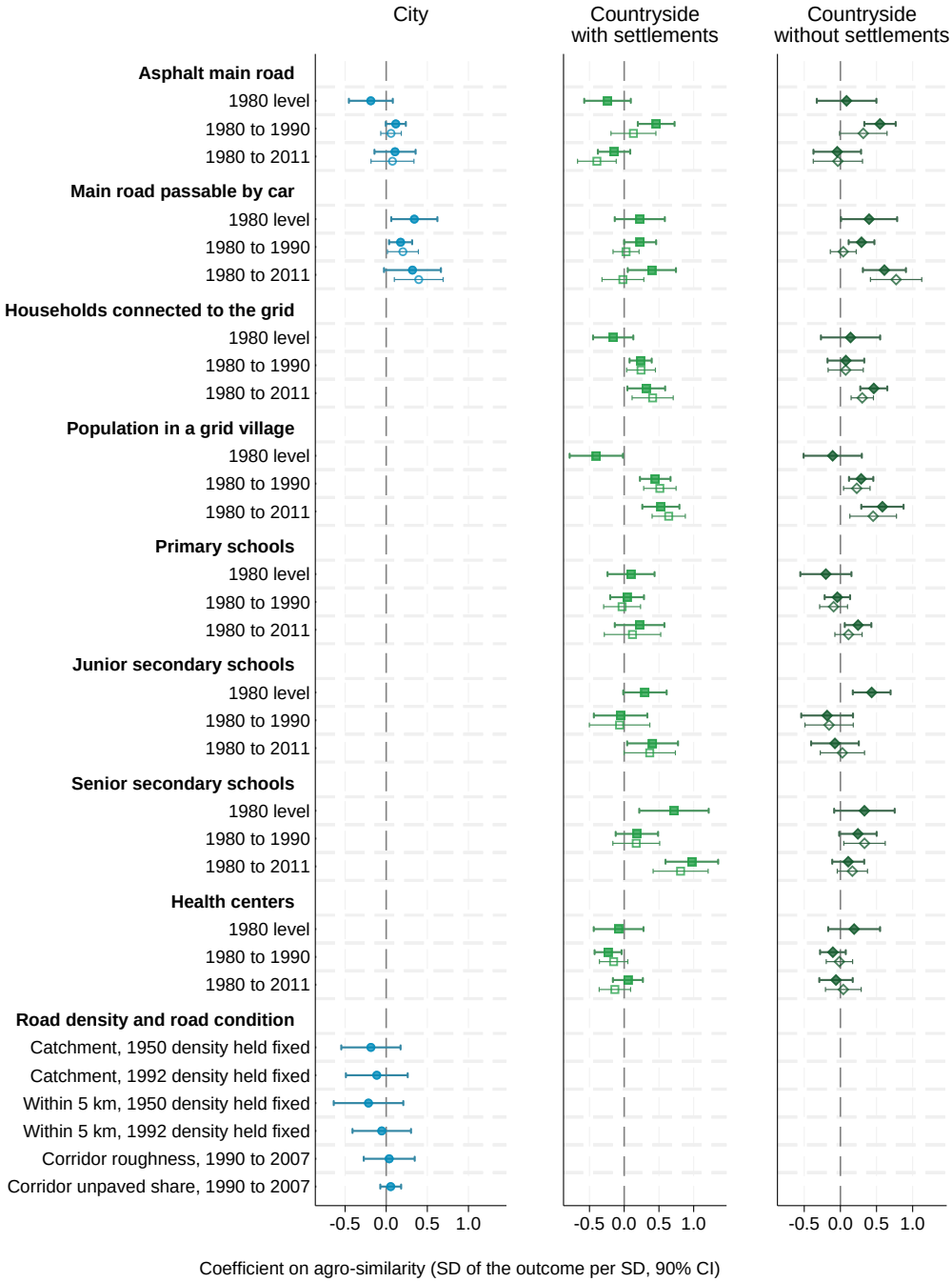
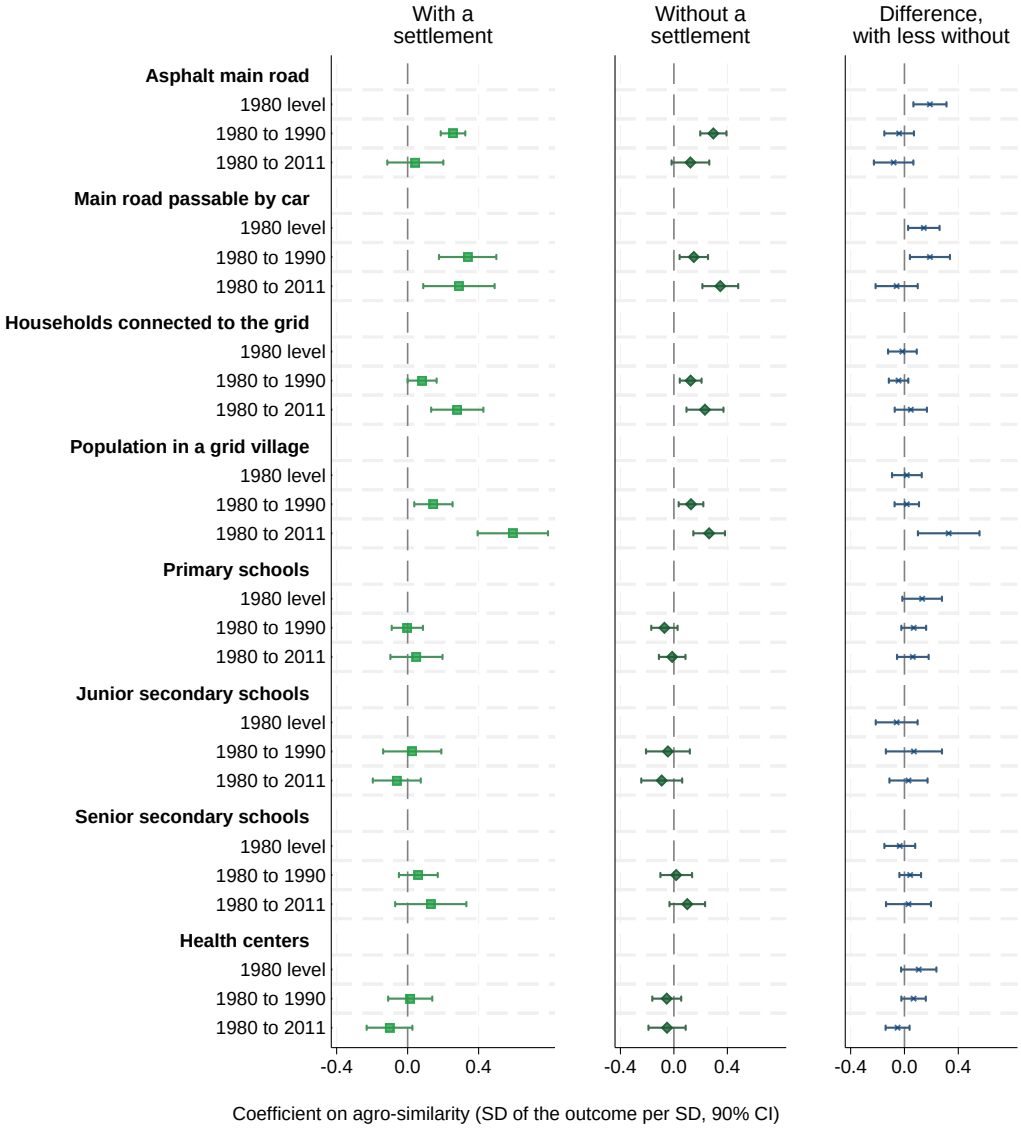


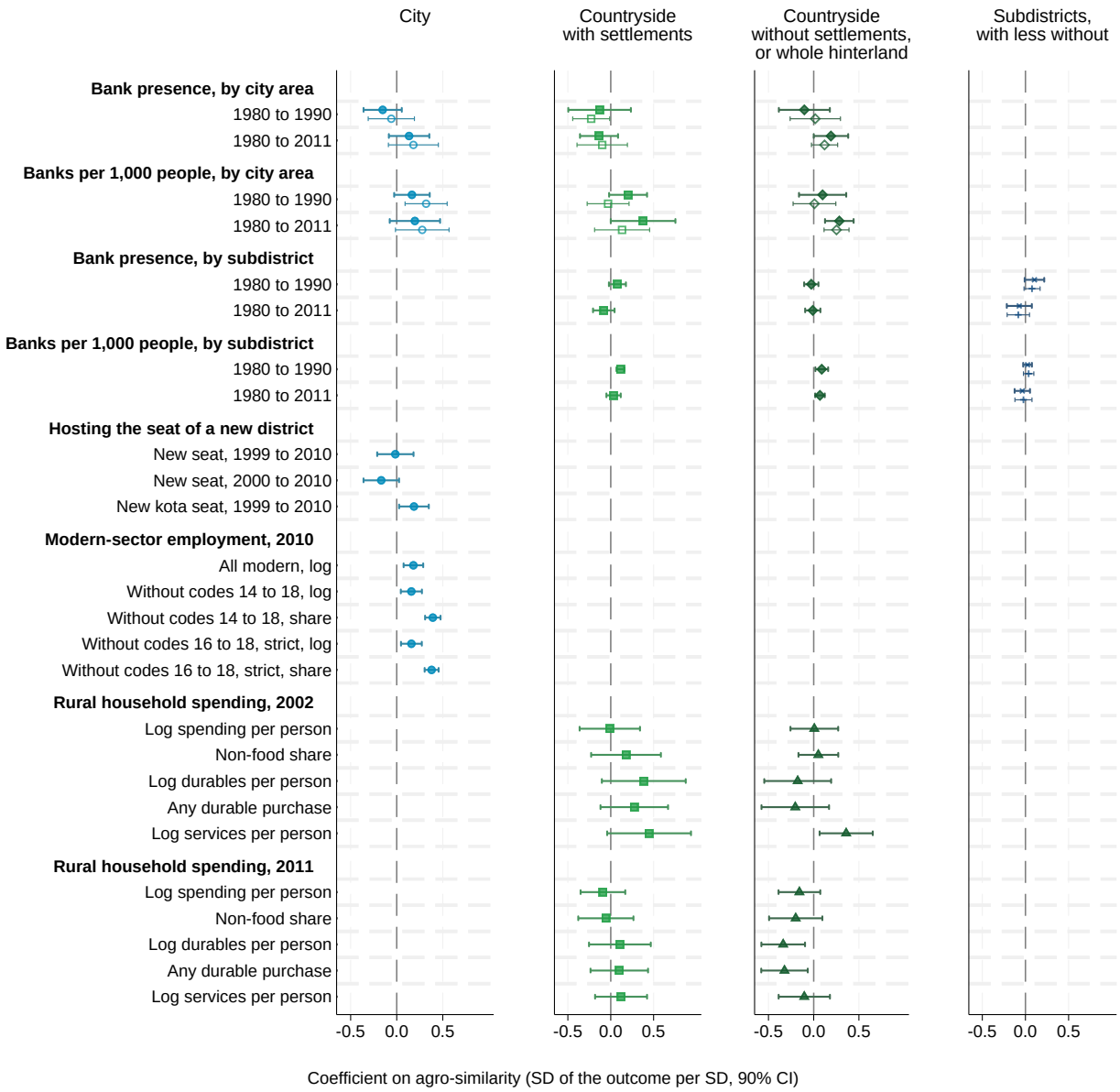
Figure S6: Public Provision and Roads in the City and Its Countryside, Continued

(b) By Rural Subdistrict



Notes: Each marker is one estimate of the appendix tests of public provision and roads: the coefficient on standardized agro-similarity divided by the SD of its regression's dependent variable over that regression's estimation sample, with its 90% interval from Conley SE within 200 km (uniform kernel). Agro-similarity is standardized as each source table standardizes it, over every city with a measured exposure in panel (a) and over the 794 rural subdistricts in panel (b). Every regression is in the main specification with the full set of controls, plus the outcome's own 1980 level (filled markers) or ln(1 + its 1980 numerator) and, in the countryside, ln(1 + its 1980 village-census population) (hollow markers). A 1980 level row regresses the 1980 value itself on exposure, without the four 1980 census baselines. Each estimate is divided by the SD of its own regression's dependent variable, so the provision rows (an endline level, or the 1980 value) and the corridor rows (a change) are scaled by different SDs. Panel (a): the main city sample (134 cities), each outcome measured over the city's member subdistricts, over the countryside with settlements (the rural subdistricts with a transmigration village within 300 travel minutes of the city) or over the countryside without settlements (the other rural subdistricts in that window). Road density on the current map holds the same density on the 1950 atlas or the 1992 map fixed, and the corridor changes from 1990 to 2007 hold their 1990 level fixed. Panel (b): the 794 in-sample rural subdistricts (191 with a settlement, 603 without), one regression per outcome and window with exposure interacted with the two indicators. For a subdistrict with a settlement the settler count is the number of settlers placed in the transmigration villages that overlap it. The difference is the slope with a settlement less the slope without. Shares are of the population (of households for the grid connections), and facilities are per 1,000 people. The 1980 to 2011 presence shares leave out West Kalimantan, and the 1980 to 2011 road shares the West Sumatra subdistricts recorded as nagari in 2011. Sources: village censuses (PODES) of 1980, 1990 and 2011, the population census of 1980, the 1950 road atlas, the current road map, and the road-condition survey of 1990 and 2007.

Figure S7: Banks, New District Seats, Modern Employment and Rural Spending



Notes: Each marker is one estimate of the appendix tests of banks, new district seats, modern employment and rural spending: the coefficient on standardized agro-similarity divided by the SD of its regression's dependent variable over that regression's estimation sample, with its 90% interval from Conley SE within 200 km (uniform kernel), in the main specification with the full set of controls. Agro-similarity is standardized as each source table standardizes it, over every city with a measured exposure for the city rows and over the 794 rural subdistricts for the subdistrict rows. Banks: the change from 1980, with the outcome's own 1980 level held fixed (filled markers) or the log of the 1980 banks plus one and, in the countryside, of the 1980 village-census population plus one (hollow markers), by area of the city on the main city sample (134 cities) and on the 794 in-sample rural subdistricts (191 with a settlement), where the difference is the slope with a settlement less the slope without. Each estimate is divided by the SD of its own regression's dependent variable, so the bank rows (a change from 1980) and the provision rows of Figure S6 (an endline level) are scaled by different SDs. Bank presence is the population share in villages with a bank, and its 1980 to 2011 change leaves out West Kalimantan and the West Sumatra subdistricts recorded as nagari in 2011. New district seats: the linear probability of hosting the seat of a district created in the window, with the city's district-seat status at the end of 1998 held fixed. Modern employment, 2010: the headline sector, and the sector without census industry codes 14 to 18 or, on the strict employment definition, 16 to 18, each with ln(1 + its own 1980 count) held fixed. Rural spending: means over the rural households of the countryside with settlements (squares) and of the whole hinterland, every rural subdistrict in the city's 300-minute window (triangles, third column), from Susenas 2002 and 2011, which start after 1980, so no 1980 level enters. Sources: village censuses (PODES) of 1980, 1990, 2011 and 2014, population censuses of 1980 and 2010, Susenas 2002 and 2011, and the World Bank district proliferation crosswalk.

S4 Additional Urban Employment Results

The census outcomes behind the headline table are reported in the three-sector levels and shares layouts that the main text consolidates into the modern-sector aggregation of Table 4 (Panels A and B of Table S9).

Table S9: Population and Sectoral Employment in 2010: Levels, Shares and Service Types

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Log population, employment and sector employment							
	Population + Employment			Log sector employment			
	ln pop	ln emp	unemp rate	L agric	L manuf	L constr	L serv
Agro-similarity	0.051	0.024	0.008	-0.166*	0.249**	0.050	0.144**
	(0.059)	(0.057)	(0.007)	(0.093)	(0.120)	(0.067)	(0.067)
Outcome mean	11.842	10.933	0.122	9.822	7.556	8.011	10.136
Sector share (2010)				0.374	0.047	0.060	0.470
Outcome SD	0.854	0.840	0.040	0.896	1.356	1.111	0.966
Observations	134	134	134	134	134	134	134
Panel B: Sectoral employment shares							
				Sectoral Employment Shares			
				agric %	manuf %	constr %	serv %
Agro-similarity				-0.051***	0.010**	-0.002	0.056***
				(0.018)	(0.005)	(0.004)	(0.018)
Outcome mean				0.374	0.047	0.060	0.470
Outcome SD				0.183	0.042	0.027	0.145
Observations				134	134	134	134
Panel C: Employment by service type							
	Service Types (Log)			Service Types (% of All Empl)			
	Trade	Transport	Skilled	Trade	Transport	Skilled	
Agro-similarity	0.114	0.135***	0.105	0.013***	0.006**	0.009*	
	(0.077)	(0.049)	(0.068)	(0.005)	(0.003)	(0.005)	
Outcome mean	9.137	7.889	8.442	0.172	0.053	0.086	
Sector share (2010)	0.172	0.053	0.086				
Outcome SD	0.999	1.057	0.915	0.051	0.024	0.028	
Observations	134	134	134	134	134	134	
Settler count	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agric Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓

Notes: Each column is one regression of the outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Standard errors are in parentheses. Panel A holds 2010 census log population, log employment, the unemployment rate (a fraction), and log employment by sector, and its sector employment share row is the share of each sector in 2010 employment. Panel B holds the share of each sector in total employment (agriculture, manufacturing, construction and services), as fractions between 0 and 1. Panel C holds 2010 census log employment in three selected service groups and their shares of total employment, as fractions between 0 and 1, and its sector employment share row is the share of 2010 employment in the service type of each column. Trade is industry 11, transport is transportation and warehousing (13), and the group labeled skilled services combines information and communication, finance and insurance, education, and health (14 to 17). These groups exclude hotels and restaurants (12) and social, government, and other services (18), so they do not exhaust services, and the grouping does not identify private ownership. Columns that would repeat an earlier panel are left out: log population, log employment and the unemployment rate in Panel B, and all services in logs and in shares in Panel C, which are column 7 of Panels A and B. Baseline rows are 1980 logs of population, total employment and agricultural employment, and Urban share 1980 is the urban share of the 1980 population. SD is the standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S4.1 Service Subsectors

Panel C of Table S9 reports employment in trade, transport, and the service group comprising information, finance, education, and health. These categories do not exhaust total service employment; their shares use total employment as the denominator.

S4.2 Manufacturing and Services

The headline estimates aggregate manufacturing and services into the modern sector. The manufacturing and services columns of Panel A of Table S9 report log employment responses of +0.249 and +0.144, respectively. Their employment-share responses are +0.010 and +0.056 in Panel B. The log coefficients describe proportional differences within each sector.

S4.3 Robustness of the Migration Bound and the Agricultural-Share Decomposition

These exhibits belong to the migration bound of Appendix E.3 and the decomposition of Appendix E.4.

Robustness. I repeat the test under the assumptions that could overturn it. Two are substantive. Exposure may shift the arrivals' own sector mix, which would lower a and with it the requirement; the test rejects at the high-exposure city's own value. Exposure is also imperfectly balanced on the 1980 agricultural share, so conditioning on that share turns Δ into a 1980-to-2010 change in the share; the test still rejects (Table S10).

Table S10: The Migration Bound Under Alternative Assumptions

	Required inflow	One-sided p
Headline: 2010 base, lifetime migrant stock, $a = 0.269$	94.0%	0.034
Five-year movers instead, $a = 0.210$	44.8%	0.025
a at the high-exposure city's own value, 0.228	53.5%	0.025
1980 agricultural share added as a control	95.4%	0.001
Both corrections together	54.1%	0.001
Conservative 1980 base, pre-treatment $a = 0.402$	47.6%	0.025
Conservative 1980 base, 2010 $a = 0.210$	17.0%	0.064

Notes: Log employment and the agricultural employment share are each regressed on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. The required inflow is the gross inflow, as a percentage of the workforce, that Equation 20 needs to produce the estimated fall in the agricultural share per standard deviation of agro-similarity when arrivals farm at share a and every incumbent stays in its sector; the base is the mean agricultural share of the sample in 2010, or in 1980 in the conservative rows. a is the agricultural share of 2010 workers born outside the district (the lifetime migrant stock) or who moved district within five years (five-year movers), or of 1980 workers born outside the province; the high-exposure city is one standard deviation higher in agro-similarity, and its a adds the estimated effect of that rise. The one-sided p, from the same standard errors, tests the null that the log employment response is at least as large as the requirement; its cell takes one shade at 0.10 or above and deepening shades below 0.10, 0.05 and 0.01.

The required composition of arrivals

The question can be pushed some way without measuring the arrivals' sector at all. Suppose every worker the exposure adds to a city works outside agriculture, the most generous assumption available to the migration account. With the incumbent workforce fixed and net employment growth equated to arrivals, a log employment response β_L implies an arrivals share $1 - \exp(-\beta_L)$ of realized employment, and at a zero agricultural share among them the city share falls by that fraction times the baseline share. At the estimated $\beta_L = 0.023$ for workers with a recorded industry and a sample mean agricultural share of 0.374 this ceiling is 0.9 percentage points, 17.1% of the estimated 5.1-point fall. An alternative version replaces the employment response with the estimated 3.5 percentage-point rise in the share of workers born outside the district and multiplies by the locally born agricultural share of 0.404, giving 1.4 points, or 28.1%.

Both point-estimate ceilings sit below the fall: under the stated assumptions, arrivals account for 28.1% of it at most. Figure S8a reports the two ceilings, the ceiling at the top of the employment response's 90% range, and the measured composition component.

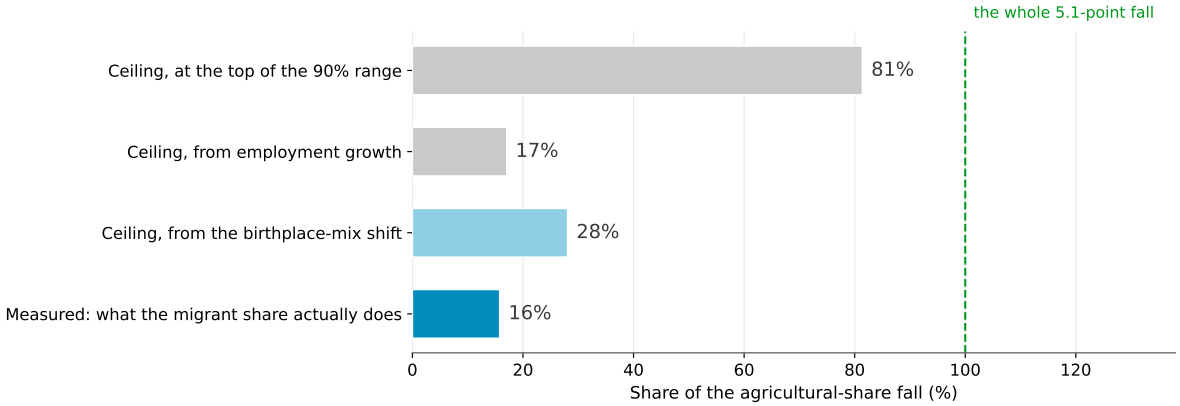
The composition channel is present but small. Migrants are markedly less agricultural than the locally born, 20 to 30% against 40%, which makes a migrant-composition account plausible before it is measured. It stays small because the heavily agricultural group is two-thirds of the workforce and exposure raises the migrant share by only a few percentage points, which cannot produce the estimated 5.1-point fall (Figure S8b).

Estimates and robustness

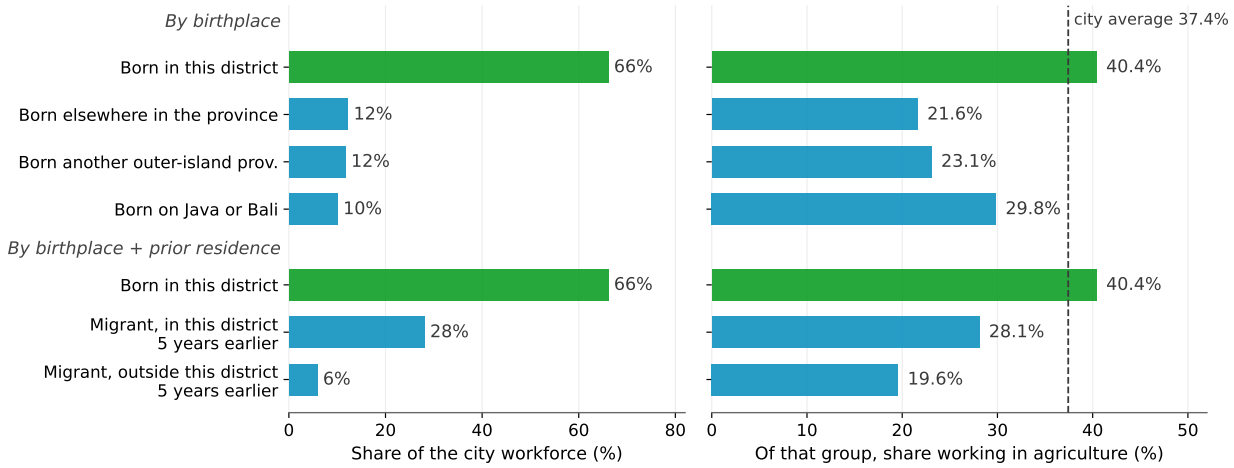
Dropping the 5 cities that draw part of their workforce from a subdistrict shared with a neighbor (Appendix E.4) gives a within share of 83.7% on the remaining 129 cities, against 83.7% on the full sample. Table S11 reports every component under both partitions, and Table S12 collects the responses, baselines, ceilings, and measured components in one place.

Figure S8: The Migration Ceilings and the Size and Agricultural Share of Each Origin Group

(a) The Migration Ceilings Against the Measured Composition Component



(b) Origin Groups: Share of the Workforce and Agricultural Share



Notes: Both panels use the main city sample and the 2010 population census; the estimates are regressions on standardized agro-similarity in the main specification with the full set of controls and Conley SE within 200 km. Panel (a): the three ceilings ask how much of the estimated agricultural-share fall migration could produce if every arrival worked outside agriculture, so they are maxima under that assumption. Each bar is a percentage of the estimated fall in the agricultural employment share per standard deviation of agro-similarity, and the dashed line at 100% is the whole fall. The employment-growth ceiling multiplies the inflow implied by the estimated log employment response by the mean 2010 agricultural share; the ceiling at the top of the 90% range uses the upper end of that response's 90% confidence interval from the same standard errors; the birthplace-mix ceiling multiplies the estimated rise in the share of workers born outside the district by the agricultural share of locally born workers. The bottom bar is the measured composition component of Figure E7. Panel (b): unweighted means across cities of each origin group's share of the workforce (left) and of its own agricultural employment share (right), with the unweighted mean city agricultural share as the dashed line. The two blocks are two partitions of the same workforce, so the shares within each block sum to one. Every entry is a mean across cities, so the group shares and the group agricultural shares do not multiply out to the city average; the gap is the covariance across cities between a group's share of the workforce and its agricultural share, summed over the groups.

Table S11: Decomposition of the Fall in the Urban Agricultural Employment Share, by Birthplace and Arrival Recency

		By birthplace			By arrival recency		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Total	Migrant share	**Within origin**	Interaction	Migrant share	**Within origin**	Interaction
Agro-similarity	-0.0509***	-0.0080**	-0.0426***	-0.0003	-0.0057	-0.0445***	-0.0007
	(0.0184)	(0.0040)	(0.0165)	(0.0055)	(0.0035)	(0.0171)	(0.0044)
Settler count	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agric Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓
Observations	134	134	134	134	134	134	134

Notes: Each column is one regression on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Standard errors are in parentheses. Column 1 is each city's 2010 agricultural employment share in the 2010 population census. Columns 2 to 4 split it exactly over four birthplace groups (born in the district, elsewhere in the province, in another outer-island province, and on Java or Bali): Migrant share, the composition component, lets only the birthplace makeup of the workforce vary; Within origin, the within component, lets only each group's own agricultural share vary; Interaction is the remainder. Columns 5 to 7 repeat the split over arrival recency: workers who never left their birth district, migrants present five years before the census, and migrants who arrived within five years. Because the census is a repeated cross-section, the within-origin component includes cohort replacement. Agro-climatic controls are the 16 village covariates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S12: Migrant Composition as a Bound on the Agricultural-Share Fall

	Estimate	Share of the fall
Panel A: Estimated responses per +1 SD of agro-similarity		
Agricultural employment share	-0.0509***	
	(0.0184)	
Conley 200 km p-value	0.006	
Log employment	0.0235	
	(0.0570)	
Conley 200 km p-value	0.681	
Share of workers born outside the district	0.0353	
	(0.0258)	
Conley 200 km p-value	0.170	
Panel B: Baseline agricultural shares, sample means		
All workers, 2010	0.374	
All workers, 1980	0.560	
Workers born in the district, 2010	0.404	
Workers who arrived within 5 years, 2010	0.196	
Panel C: Maximum composition effect, arrivals assumed entirely non-agricultural		
Inflow based, 2010 baseline	0.0087	17.1%
at the 90% upper end	0.0414	81.4%
on the 1980 baseline	0.0130	25.5%
Composition based, 2010 baseline	0.0143	28.1%
at the 90% upper end	0.0314	61.7%
Panel D: Measured, replacing the assumption with the observed arrival mix		
Migrant share, by birthplace	-0.0080**	15.8%
Within origin, by birthplace	-0.0426***	83.7%
Migrant share, by arrival recency	-0.0057	11.1%
Within origin, by arrival recency	-0.0445***	87.5%

Notes: Panels A and D report regressions on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Standard errors are in parentheses, with the p-value beneath each, shaded by significance level. Workers are counted in the 2010 population census, and in the 1980 census for the rows marked 1980; Panel B gives sample means. Panel C bounds the composition effect by assuming every arrival works outside agriculture: the inflow-based bound multiplies the implied employment inflow, one minus the exponential of minus the log employment response, by the baseline agricultural share; the composition-based bound multiplies the estimated rise in the share of workers born outside the district by the agricultural share of locally born workers; the upper-end rows use the upper end of the response's 90% confidence interval; the larger bound, here the composition based one, is the ceiling. Panel D replaces the assumption with the arrivals' measured agricultural share, from Table S11. Share of the fall divides each Panel C bound by the Panel A fall in the agricultural employment share, and each Panel D component by the total it decomposes. SD is one standard deviation of agro-similarity, and Log employment is the log of the city's 2010 employment. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S5 Food Prices, Consumption Composition, and Trade Openness

S5.1 Food-Price Indices and Rice Unit Values

The 1996 food-price basket estimates are statistically insignificant under each weighting: the raw-food index is −0.012 (standard error 0.025) with city-specific expenditure weights, −0.012 with pooled urban-center weights, and −0.012 equal-weighted, and the all-food index including prepared food is +0.007. The rice unit-value estimate is also statistically insignificant. Both the 1996 and the 2002 waves predate the 2004 rice-import ban, whose openness-differential price effects (Bazzi 2017) run through the 2011 wave, and the 1996 wave includes more of the main city sample than the 2002 wave (Table A5). Panel A of Table S13 and Appendix Table E4 report these food-price estimates.

Table S13: Food-Price and Rice Unit-Value Responses to Agro-Similarity, 1996 and 2002

Panel A: The 1996 raw-food price index under two other weightings					
	Raw-ag Laspeyres	Raw-ag eq-wt			
Agro-similarity	-0.012	-0.012			
	(0.024)	(0.025)			
Outcome mean	1.041	1.040			
Outcome SD	0.115	0.135			
N (cities)	112	112			
Panel B: The raw-food price index in levels and changes, 1996 to 2002					
	Log price 1996	Log price 2002	Change 1996 to 2002		
Agro-similarity	-0.004	0.013	0.017		
	(0.013)	(0.011)	(0.011)		
p-value	0.759	0.226	0.134		
Observations	102	102	102		
Panel C: The 1996 rice unit value by quality					
	ln rice (pooled)	ln local rice	ln high-qual rice	HQ spend share	HQ buyer share
Agro-similarity	-0.009	0.031*	0.056	0.193***	0.182***
	(0.029)	(0.017)	(0.068)	(0.061)	(0.063)
Outcome mean	6.840	6.845	6.833	0.268	0.262
Outcome SD	0.136	0.130	0.150	0.297	0.289
N (cities)	112	97	49	112	112
Settler count	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓
Agric Empl 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓

Notes: Each column is a separate urban-center regression on standardized agro-similarity, in the main specification with the full set of controls and Conley SE within 200 km. Check marks show the controls. Standard errors are in parentheses. Panel A covers the 112 urban centers of the main city sample with food prices in the 1996 national socioeconomic household survey (Susenas). Its two columns are a price index of raw agricultural foods (cereals to spices), the median unit value of each item in the urban center relative to its median over the households of all urban centers, averaged with spending weights pooled over all urban centers (column 1) or as an equally weighted geometric mean (column 2). The same index with the spending weights of the urban center, the index over every food from cereals through prepared food, and the log rice unit value are in Appendix Table E4. Panel B covers the 102 urban centers of the main city sample with an agricultural price index in both the 1996 and 2002 household-survey (Susenas) waves. Its coefficients are in log units of one plus the index, and the p-values come from the same inference. The outcome is the log of one plus the raw-food Paasche price index, in 1996 (column 1), in 2002 (column 2), and as the 2002 log minus the 1996 log (column 3). Panel C covers the urban centers of the main city sample where the 1996 household survey (Susenas) measures each outcome. Column 1 is the log unit value of rice (expenditure over quantity, in rupiah per kilogram), local and high-quality varieties pooled. Columns 2 and 3 are the log unit values of local and of high-quality rice. Column 4 is the high-quality share of rice expenditure and column 5 the share of rice-buying households that buy any high-quality rice, both fractions. Ln is the natural log and HQ is high-quality. The Baseline Pop 1980, Total Empl 1980 and Agric Empl 1980 rows are 1980 baselines in logs (population, employment and agricultural employment). SD is the standard deviation, and N (cities) is the number of urban centers in each regression. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S5.2 Price Changes on the Common-City Sample

Both the 1996 and the 2002 waves include the agricultural price index, so on the 102 urban centers that appear in both it can be differenced, and exposure can be asked whether it moved the price over the intervening period as well as where the price stood at a point in time. The estimates for the change and both cross-sections are statistically insignificant. Differencing removes a time-invariant city price component. The interval ends before the 2004 rice-import restrictions, and the index is a unit value,

which moves with the quality and the composition of the basket as well as with the price of a fixed good (Panel B of Table S13).

S5.3 Rice Quality and Expenditure Composition

The 1996 module splits rice into local and high-quality categories, allowing a comparison of their unit values and expenditure shares. The categories do not hold quality or household selection fixed. Households shift toward the high-quality variety: its share of rice spending is 19.3 percentage points higher per standard deviation of exposure, from a mean of 26.8%, and the share of rice buyers who buy any of it rises with it, both statistically significant. The local-variety unit-value estimate is positive and marginally statistically significant. The high-quality and pooled estimates are not statistically significant. In 1996, then, more households buy higher-quality rice with a pooled unit-value response that is not statistically significant (Panel C of Table S13).

S5.4 Trade Openness by Location Type

The paper's figure of the response by location type split on trade openness (Figure 18) shows a star beside each gap between the more closed and the more open group of a location type whose p-value under the figure's Conley standard errors is below 10%. Table S14 reports, for each of its rows, how many units of the location's sample the outcome covers, the estimation rows in each group, the response of the more open group less that of the more closed group with its standard error, and the p-value of that gap under the Bartlett kernel at the same cutoff and with clusters by province. Population, employment and the unemployment rate are from the 2010 census; the real wage, the food spending share and spending per person are from the 2011 wave of the household survey (Susenas).

Table S14: The Openness Split of the Response by Location Type: Coverage and the Test of the Difference

Outcome	Location	Measured of sample	Observations Closed	Observations Open	Open minus closed	*p*-value of the difference Conley	Bartlett	Province
Panel A: Population and employment								
Log population	Cities	134 / 134	66	68	−0.150	0.145	0.186	0.160
					(0.103)			
	No settlement	603 / 603	305	298	0.147	0.118	0.101	0.214
					(0.094)			
	With settlement	191 / 191	96	95	−0.240	0.263	0.202	0.240
					(0.214)			
Log employment	Cities	134 / 134	66	68	−0.149	0.199	0.217	0.179
					(0.116)			
	No settlement	603 / 603	305	298	0.165*	0.052	0.038	0.128
					(0.085)			
	With settlement	191 / 191	96	95	−0.312	0.177	0.125	0.176
					(0.231)			
Log agricultural employment	Cities	134 / 134	66	68	0.079	0.642	0.608	0.629
					(0.170)			
	No settlement	603 / 603	305	298	0.086	0.339	0.341	0.391
					(0.090)			
	With settlement	191 / 191	96	95	−0.338*	0.095	0.056	0.102
					(0.203)			
Log modern employment	Cities	134 / 134	66	68	−0.248**	0.027	0.033	0.061
					(0.112)			
	No settlement	603 / 603	305	298	0.135	0.204	0.198	0.250
					(0.107)			
	With settlement	191 / 191	96	95	−0.182	0.401	0.357	0.392
					(0.216)			
Panel B: Wages, unemployment and spending								
Log real wage	Cities	134 / 134	66	68	0.074	0.628	0.691	0.705
					(0.152)			
	No settlement	580 / 603	295	285	−0.094	0.387	0.492	0.515
					(0.109)			
	With settlement	188 / 191	95	93	0.012	0.959	0.958	0.959
					(0.229)			
Unemployment rate	Cities	134 / 134	66	68	−0.239	0.409	0.378	0.461
					(0.289)			
	No settlement	603 / 603	305	298	0.016	0.934	0.926	0.935
					(0.191)			
	With settlement	191 / 191	96	95	0.209	0.161	0.176	0.263
					(0.149)			
Food spending share	Cities	134 / 134	66	68	0.141	0.382	0.384	0.532
					(0.161)			
	No settlement	582 / 603	296	286	−0.290	0.141	0.092	0.254
					(0.197)			
	With settlement	188 / 191	95	93	0.587***	0.006	0.011	0.024
					(0.215)			
Log spending per person	Cities	134 / 134	66	68	−0.041	0.736	0.798	0.855
					(0.120)			
	No settlement	582 / 603	296	286	0.229**	0.021	0.046	0.092
					(0.099)			
	With settlement	188 / 191	95	93	−0.365**	0.025	0.037	0.030
					(0.163)			

Notes: The rows of Figure 18: one regression per outcome and location type, on the units of that location's sample that report the outcome, with each location type split at the median route distance to the Thai and Vietnamese rice hubs. Open minus closed: the response of the more open group less that of the more closed group, in standard deviations of the outcome per standard deviation of agro-similarity, with its Conley SE within 200 km (uniform kernel) beneath; its stars are the figure's. The *p*-values test that gap under those Conley SE, the Bartlett kernel at the same cutoff, and province clusters; a cell takes one shade at 0.10 or above and deepening shades below 0.10, 0.05 and 0.01. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S6 Urban Emergence and Endline Sample Selection

The set of cities that defines the estimation sample uses the GHS Urban Center Database's fixed 2025 boundaries, so the timing defense in Section 4.3 requires knowing whether the shock itself created urban

units that a 2025-defined sample would then select. This appendix reports that test directly, estimating the effect of the same exposure on urban emergence outside the fixed sample.

S6.1 Lattice Sample and Historical Satellite Inputs

The unit is an equal-area 25 km lattice over outer-island land (2,648 cells, excluding Java, Bali, and Papua as in the main sample), with the pre-shock urban baseline taken from the GHS-SMOD 1975 classification, the published 1975 classification. The treatment is the paper's exposure rebuilt at cell centroids, the unweighted mean of village agro-climatic similarity within 300 route-aware cost-minutes, with the agro-climatic control set and the transmigrant-count control re-derived from the same cell-to-village pairs and Conley standard errors at a 200-kilometer cutoff for the lattice regressions. The estimation sample is the baseline-rural, exposure-defined cells containing no transmigrant settlement point (program settlement points are excluded): 572 cells at 25 km. New-town founding is a rare event (single-digit foundings in the exposed sample by 2010), so the founding analyses use the finer 15 km lattice (2,074 cells), defining a founding as a nightlights cluster disjoint from the 1992 urban extent with a 5 km guard against an existing city's expanding edge. A town that formed between settlement (1979 to 1988) and 1992 is part of the 1992 urban extent, so the founding count covers towns formed after 1992.

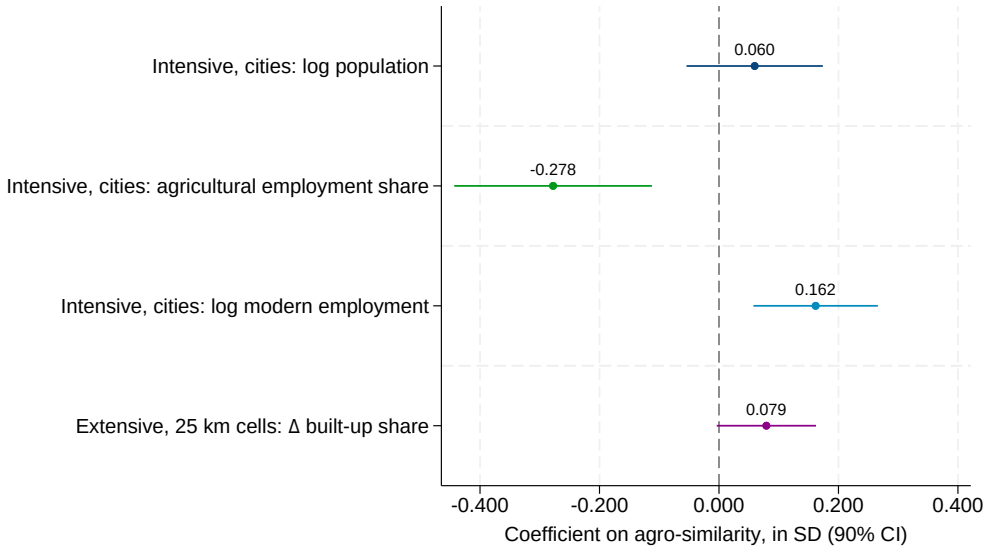
The early classification, built-up share, and gridded population depend partly on interpolated satellite inputs. The historical coverage limitations in Section S2 therefore also apply to the sample definition and baseline controls here. The exercise does not establish the direction or size of the resulting bias.

S6.2 Built-Up Growth and Urban Founding

On one standardized axis, the extensive response is smaller than the intensive responses of the agricultural employment share and modern employment, and it is not statistically significant (Figure S9). The intensive trio sits off zero on its two composition outcomes (ln population +0.060, agricultural employment share −0.278, ln modern employment +0.162), while the extensive response, the 1975 to 2010 change in built-up areal share, is smaller at +0.079, its 90% interval crossing zero. Panel A of Table S15 reports the raw-units version: +0.00011 of built-up areal share per standard deviation of exposure over 1975 to 2010 (SE 0.00007), stable across the 1975 to 2000 and 1975 to 2020 windows. The estimate is sensitive to geographic controls. Panel B of Table S15 shows the same coefficient conventionally significant within islands (island fixed effects +0.00017, province fixed effects +0.00016, both statistically significant). The additional tables vary the baseline and founding definitions. Panel C of Table S15 reruns the headline under the other baseline-rural definitions and Panel A of Table S16 the founding event under all four founding definitions, positive throughout.

The nearby-exposure comparison examines spatial redistribution. Panel B of Table S16 adds the mean exposure of cells 30 to 100 km away: the own effect stays positive while the ring enters negative (founding −0.0034, marginally significant), consistent with local displacement. The ring's

Figure S9: Intensive Versus Extensive Margin of the Agro-Similarity Shock



Notes: Each marker is a coefficient, printed above it, in SD of the outcome per SD of agro-similarity; bars are 90% confidence intervals from Conley SE within 200 km. The first three rows are 2010 log population, agricultural employment share and log manufacturing and services employment on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline; 134 cities), in the main specification with the full set of controls. The fourth is the 1975-2010 change in the built-up share of land, from the Global Human Settlement Layer, across 572 square 25-kilometer cells of outer-island land outside Papua that were rural in that source's 1975 settlement map, lie within 300 travel minutes of a transmigration village and contain none; its controls are the cell's settler count and 16 agro-climatic covariates within 300 minutes, its 1975 built-up share, log 1975 gridded population and log land area.

precision is sensitive to the band definition, so it is illustrative; the magnitude statements that make the displacement point are in checkable units. The count of founded towns per district does not rise with exposure (Panel C of Table S16), and the population bridge in Table S17 reports a positive but imprecise population association within incumbent footprints and a negative association within newly founded footprints (−1,050 per standard deviation, statistically significant).

The built-up, founding, and district-population comparisons use different outcomes and geographic units. Their standardized coefficients do not measure contributions to aggregate urban growth. Sparse founding events, imprecise population estimates, and the early-satellite limitations qualify the interpretation.

S6.3 Sensitivity to the City Inclusion Threshold

One source of selection is left over, and the lattice cannot test it: the estimation sample above is restricted to baseline-rural cells, which excludes precisely the incumbent towns that sat near the inclusion threshold. Membership in the urban-center sample requires 50,000 people in the 2025 population grid, a post-treatment quantity, so an incumbent town near that line enters only if it grew, and exposed marginal towns would select in while unexposed ones select out. Of the 134 sample cities, 18 have a 2025 polygon population within 20% of the threshold, and the sample's smallest is 50,287, close enough

to the line that the concern is not hypothetical. Dropping all 18 and reestimating Table 4 does not move the population coefficient in the direction the concern predicts. The agricultural employment share coefficient goes from -0.051 to -0.044 ($p = 0.031$), modern-sector employment from 0.158 to 0.152 ($p = 0.038$), and log population from 0.051 to 0.082 ($p = 0.234$, against $p = 0.389$ on the full sample), all on the remaining 116 cities. The population estimate remains statistically insignificant without the near-threshold cities. This exclusion is a sensitivity check; it does not identify the direction of selection bias or exclude selection in the sectoral-share estimates. Two further sample cities have no GHS polygon population because the reclassification created them as new nodes instead of selecting them on the threshold, so they are not exposed to this selection and are retained.

Table S15: Built-Up Share Changes on Agro-Similarity

	Δ built-up	N
Panel A: Windows of the change, rural by the 1975 settlement map, no fixed effects		
1975 to 2010	0.00011	572
	(0.00007)	
1975 to 2020	0.00012	572
	(0.00008)	
1975 to 2000	0.00009*	572
	(0.00005)	
Panel B: Geographic fixed effects, 1975 to 2010		
Island fixed effects	0.00017***	572
	(0.00006)	
Province fixed effects	0.00016***	572
	(0.00006)	
Panel C: Baseline rural definitions, 1975 to 2010		
SMOD $\geq$ 23	0.00019*	727
	(0.00011)	
SMOD $\geq$ 30	0.00008	883
	(0.00013)	
Built-up 1975	0.00021	927
	(0.00014)	
NTL 1992	0.00010	906
	(0.00012)	

Notes: Each row is one least-squares regression of the change in the built-up share of each cell (Global Human Settlement Layer) from 1975 to the end year named on agro-similarity of the transmigration villages within 300 travel minutes, standardized over all cells with a measured exposure, with its number of cells in the second column. The sample is the square 25-kilometer cells of outer-island land outside Papua that count as rural at baseline and have a transmigration village within 300 travel minutes and none inside the cell. Panel A varies the end year, with no urban pixel in the 1975 settlement map of that source (class 21 or above) as the rural rule, and its first row, the change from 1975 to 2010, is the specification the other two panels vary. Panel B adds island fixed effects or province fixed effects. Panel C changes the rural rule, and a stricter urban threshold admits more cells: no pixel of the 1975 settlement map at 23 or above (dense urban cluster) or at 30 (the urban-center class), a 1975 built-up share below 3%, or no urban nighttime-lights pixel in 1992. Controls: the settler count and 16 agro-climatic covariates within 300 minutes, and the 1975 built-up share, log 1975 population and log land area of the cell. Conley SE within 200 km in parentheses. SMOD is the settlement map of the Global Human Settlement Layer, the number after it is the class threshold, and NTL is nighttime lights. Δ built-up is the change in the built-up share, agro-similarity is the standardized agro-climatic similarity of the transmigration villages, and N is the number of cells. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S16: New Urban Settlement on Agro-Similarity

	(1)	(2)	(3)	(4)
Panel A: New urban settlement under alternative definitions				
	New cluster 2010	Any urban 2010	New cluster 2020	Any urban 2020
Agro-similarity	0.00180	0.00086	0.00190	0.00140
	(0.00132)	(0.00108)	(0.00120)	(0.00142)
N	2074	2074	2074	2074
Events	6	8	7	12
Panel B: Exposure in surrounding areas				
	New town	Δ urban share		
Agro-similarity	0.0047***	0.0001***		
	(0.0015)	(0.0000)		
Ring term (30-100 km)	-0.0034*	-0.0001		
	(0.0021)	(0.0001)		
N	2074	2074		
Panel C: District-level new-cluster counts				
	Poisson	Poisson +islFE		
Agro-similarity (district)	0.3807	0.7321		
	(0.4275)	(0.5929)		
N	129	129		
FE	none	island		

Notes: In Panel A, each column is a linear probability model across square 15-kilometer cells of outer-island land outside Papua with no urban pixel in the 1975 Global Human Settlement Layer settlement map, a transmigration village within 300 travel minutes and none inside the cell. In the new-cluster columns the outcome is one when the cell overlaps an urban nighttime-lights cluster of the year in the column heading that does not touch the 1992 urban extent and has its center at least 5 kilometers from it. In the any-urban columns it is one when the cell holds any urban nighttime-lights pixel in that year. The Events row counts sample cells with an outcome of one. In Panel B, each column is a regression across the 2,074 cells, on a 15-kilometer lattice over the outer islands, that had no urban settlement in the 1975 Global Human Settlement Layer, lie within 300 travel minutes of a transmigration village and do not contain one. Column 1 is 1 when the cell overlaps a new town, a 2010 night-light urban cluster disjoint from the 1992 urban extent with its center at least 5 kilometers from it, and column 2 is the 1992 to 2010 change in the night-light urban share of the land area of the cell. The regressors are the agro-similarity of the cell, the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, and the ring term, the mean agro-similarity of the cells with a defined exposure whose centers lie 30 to 100 kilometers away. Both are standardized over the lattice cells with a defined exposure, and a negative ring coefficient means that exposure nearby suppresses founding. In Panels A and B, the regressor is standardized over the cells with a measured exposure, the controls are the settler count and 16 agro-climatic covariates within 300 minutes, and the 1975 built-up share, log 1975 population and log land area of the cell, and Conley SE within 200 km are in parentheses. In Panel C, each column is a Poisson regression, across the 129 outer-island districts with at least one at-risk cell, of the number of new towns whose centers fall in the at-risk cells of the district on the district mean of standardized cell agro-similarity, with the log number of at-risk cells as offset. Column 2 adds island-group fixed effects (Sumatra, Kalimantan, Sulawesi and Maluku). Standard errors clustered by province are in parentheses. At-risk cells are 15-kilometer lattice cells with no urban settlement in the 1975 Global Human Settlement Layer that lie within 300 travel minutes of a transmigration village and do not contain one. A new town is a 2010 night-light urban cluster outside the 1992 urban extent whose center lies at least 5 kilometers from it. Agro-similarity is the standardized agro-climatic similarity of the transmigration villages, Agro-similarity (district) is the district mean of standardized cell agro-similarity, Δ urban share is the 1992 to 2010 change in the night-light urban share, FE is fixed effects, islFE is island-group fixed effects, and N is the number of cells in Panels A and B and of districts in Panel C. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S17: Population Bridge: Persons Added per SD of Agro-Similarity, by Footprint Type

	(1) d pers incumbent	(2) d pers new town	(3) incumb 90-20	(4) new 90-20
Agro-similarity	1,346	-1,050**	2,269	-1,465**
	(1,223)	(438)	(2,338)	(620)
N	150	150	150	150
Footprint	incumbent	new town	incumbent 90-20	new town 90-20

Notes: Each column regresses the change in population, in persons, inside one footprint type of a district on district agro-similarity, standardized over the districts, across the 150 outer-island districts with a 15-kilometer lattice cell within 300 travel minutes of a transmigration village. Population is summed from the Global Human Settlement Layer population grid inside incumbent footprints, the 1992 night-light urban extent (columns 1 and 3), or new-town footprints, the 2020 night-light urban clusters outside that extent with centers at least 5 kilometers from it (columns 2 and 4); the change runs from 1990 to 2010 in columns 1 and 2 and from 1990 to 2020 in columns 3 and 4. District agro-similarity averages, over the lattice cells of the district, the mean agro-climatic similarity of the transmigration villages within 300 travel minutes of each cell. Standard errors clustered by province in parentheses. In the labels, d pers is the change in population in persons, incumb and new are the incumbent and new-town footprints, 90-20 is the change from 1990 to 2020, agro-similarity is the standardized district agro-climatic similarity of the transmigration villages, and N is the number of districts.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

S7 Service Flows and Travel-Time Gradients

The trade-cost elasticity of traded non-agricultural output, δ_S (δ_M in Table 7), is not disciplined by the moments of the estimation. This appendix measures it directly.

Trade costs for traded non-agricultural output in the model are $\tau_{ojM} = \max(d_{oj}, 1)^{\delta_S}$, entering CES demand as $\tau^{1-\sigma}$ with $\sigma = 4$, so bilateral service expenditure falls in travel time as $d^{-(\sigma-1)\delta_S}$. A gravity regression of service flows on log travel time therefore recovers $-(\sigma-1)\delta_S$ and nothing finer: σ and δ_S are not separately identified from flows, but their product is, and it is the product that governs how far a city's service market reaches.

S7.1 Survey Coverage and Located Service Records

The Indonesia Family Life Survey records the provider's location and the respondent's residence for outpatient use in the four weeks before interview. Pooling the 2007 and 2014 waves gives 9,275 records with both locations identified. Of these, 1,925 link different year-2000 subdistricts, across 1,074 positive provider and residence pairs. The descriptive shares in Figure S10 instead use the survey's same-subdistrict indicator and its survey-year boundaries. Estimating

$$F_{oj} \;=\; \exp\!\big(\mathrm{FE}_o + \mathrm{FE}_j + \beta \ln d_{oj}\big)$$

by Poisson pseudo-maximum likelihood with origin and destination fixed effects, zeros retained, and standard errors two-way clustered on origin and destination, on the route-aware cost-distance surface in minutes, gives $\hat{\beta} = -3.272$ with a standard error of 0.100, and hence

$$\hat{\delta}_S \;=\; -\hat{\beta}/(\sigma-1) \;=\; 1.091 \quad (0.033),$$

with a 90% interval of $[1.036, 1.146]$.

The estimate is similar when flows are weighted by out-of-pocket expenditure (1.131), to excluding traditional practitioners (1.133), to keeping only formal providers (1.117), and to excluding the public health centers that Indonesia sites in every subdistrict by administrative design (1.072). Restricting the sample to the outer islands, which is the estimation footprint of the paper, gives 0.948.

S7.2 Comparisons Across Services

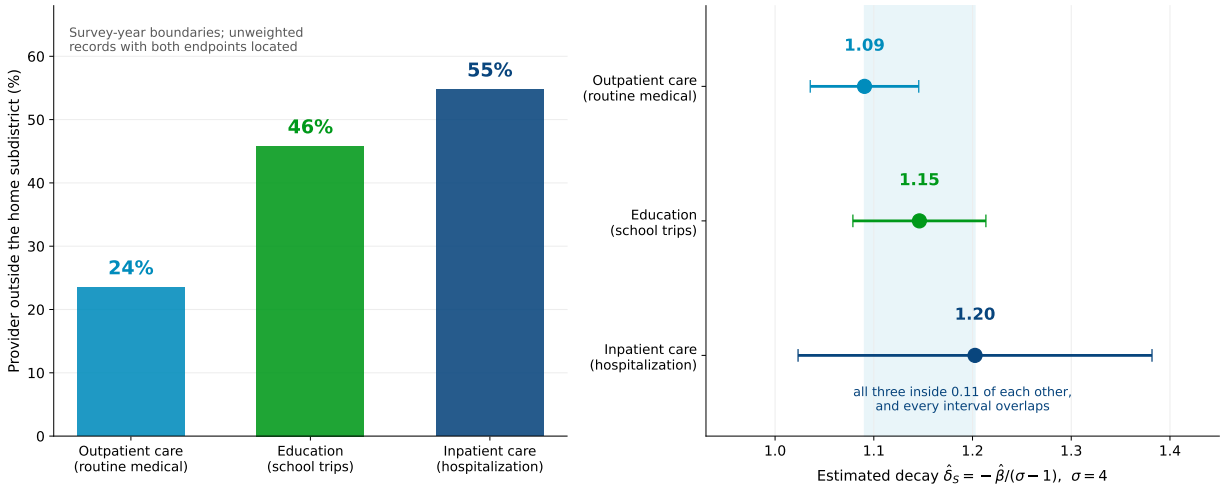
Outpatient visits are one service, and medical care is atypical in being urgent and in having a publicly sited provider network. The survey records the same provider-location block for two further services, providing further comparisons. The additional services are schooling and hospitalization. Figure S10 estimates all three identically.

The provider-location shares differ across services, while the estimated distance gradients have overlapping confidence intervals. The share of records with a provider outside the home subdistrict more than doubles from routine medical care to hospitalization, while the estimated decay moves by 0.11 and every interval overlaps every other. These are different objects: the descriptive share records the provider location relative to the residence, while the regression estimates a conditional travel-time gradient. Figure S10 separates the two.

The recorded provider location does not always describe a journey by the household: records marked as provider home visits are retained. The services covered here also exclude retail and restaurant purchases. The survey covers thirteen provinces, so the gradient is estimated on that geographic coverage and extrapolation beyond it requires an additional assumption.

The calibrated value. The measured value is 1.091 and the calibrated model is solved at $\delta_S = 1.0$, inside the outer-island interval reported in Table 7. No average of the three services in Figure S10 is used. The estimates are not independent draws, sharing households, geography and one cost surface.

Figure S10: Service Locations and Estimated Travel-Time Gradients



Notes: The left panel reports providers outside the respondent's home subdistrict under survey-year boundaries: 23.5% of records for outpatient care (routine medical), 45.8% for education (school trips) and 54.8% for inpatient care (hospitalization). The right panel reports the estimated distance decay $\hat{\delta}_S$ under year-2000 boundaries: 1.091 for outpatient care ($\hat{\beta} = -3.272$, standard error 0.100), 1.146 for education ($\hat{\beta} = -3.439$, standard error 0.123) and 1.203 for inpatient care ($\hat{\beta} = -3.608$, standard error 0.327), with 90% intervals of [1.036, 1.146], [1.079, 1.214] and [1.023, 1.382]. The descriptive shares are unweighted, using survey-year subdistrict boundaries among records with both residence and provider located: 9,275 outpatient, 4,719 education, and 699 inpatient records, including own-subdistrict observations. Provider home-visit records are retained, so provider location does not always represent household travel. The gravity regressions use year-2000 boundaries and the product of observed provider and residence locations, retain bilateral zero flows, and exclude own-subdistrict pairs and locations with no observed cross-subdistrict flow. Poisson pseudo-maximum likelihood with origin and destination fixed effects, zeros retained, standard errors two-way clustered on origin and destination, own-subdistrict pairs excluded, estimated on the model's route-aware cost distance in minutes. $\hat{\delta}_S = -\hat{\beta}/(\sigma-1)$ at $\sigma = 4$. Outpatient care and education pool the 2007 and 2014 waves, and inpatient care uses 2007 only, because the 2014 module records the province and subdistrict of the hospital but not the district. Education covers respondents enrolled at their highest level, which in the located sample is predominantly senior secondary and tertiary.

S8 Urban-Center-Level Empirical Facts and Model Coverage

Each urban center-level empirical fact maps to an object of the model, and the estimate behind it comes from the main text wherever a main-text table reports it. The population, employment, and sectoral-composition block (F1-F6) is reported here from the main-text Table 4, which consolidates the same census regressions that Appendix Section E.1 reports in a disaggregated layout. Most of the remaining Susenas- and PODES-based rows are likewise reported in the main text alongside the results they support; only the rows without a main-text table of their own (the 1996 household-survey table behind F9, the agricultural-goods outcomes behind F10, and the service-sector composition behind F20) point to Section E.1, where their signs and significance are verified on the same sample. Sample sizes below 134 in those tables follow the narrower Susenas and PODES coverage of the same sample. Table S18 lists the facts and their mapping to the model.

The structural model is calibrated to match the exact standardized (z-on-z) point estimates, reported together with the untargeted over-identifying moments (Appendix Table B5).

Table S18: Empirical Facts and Model Coverage

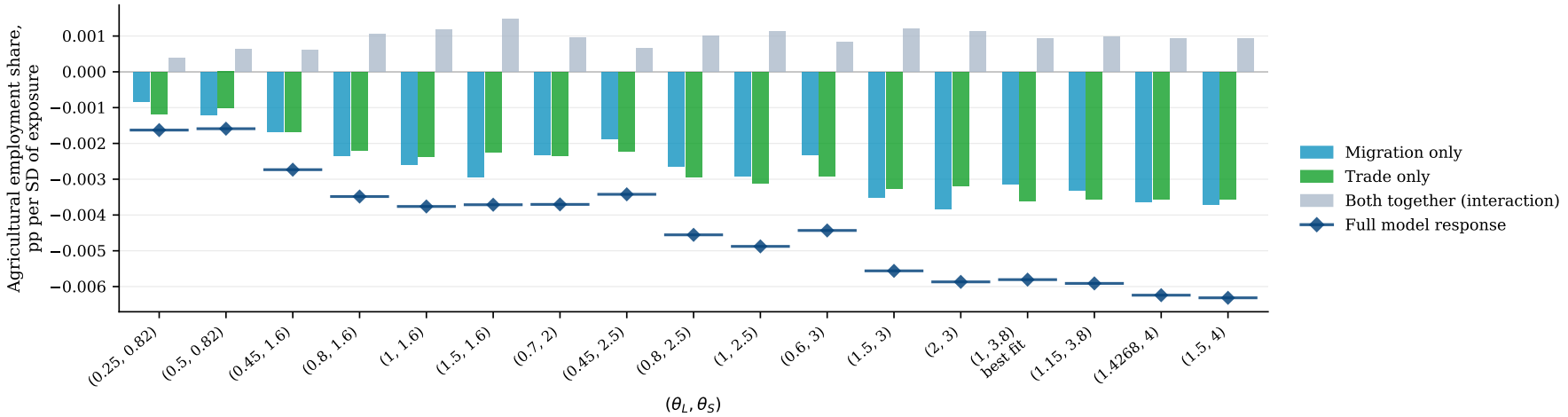
#	**Fact**	**Coef (SE)**	**Source**
F1	City population: positive, imprecise	0.051 (0.059)	Table 4 col 1
F2	City employment: positive, imprecise	0.024 (0.057)	Table 4 col 2
F3	Agricultural employment share ↓	-0.051^{***} (0.018)	Table 4 col 5
F4	Log agricultural employment ↓	-0.166^{*} (0.093)	Table 4 col 4
F5	Log modern employment ↑ (serv+manuf)	0.158^{**} (0.062)	Table 4 col 6
F6	Modern employment share ↑	0.066^{***} (0.015)	Table 4 col 7
F8	Non-food spending share ↑	Positive, sig.	Table E4 Panel B col 1
F9	Housing rent: 1996 positive, imprecise; 2011 flat	Pos., ns (1996); ≈ 0, ns (2011)	Appendix Table E4 Panel A; Panel B col 2
F10	Rice surplus ↓, imprecise; own production ↑	-0.141 (1.208); 0.736^{***}	Appendix Table E3 Panel C col 2; Panel A col 2
F12	Non-ag wage: ns	≈ 0, ns	Table E4 Panel B col 3
F15	Food spending share ↓	Negative, sig.	Appendix Table B5, food expenditure share row
F17	Village retains ag workers	Positive, ns	Figure 14
F19	Migration from the same province ↑; other outer-isl provinces ns; Java/Bali ↓	+0.042 (0.018), sig. at 5%; +0.018 (0.021), not sig.; −0.018 (0.011), sig. at 10%	Appendix Table C1 Panel B, col 1
F20	Service growth concentrated in transport ↑	Positive, sig.	Table S9 Panel C col 2

Notes: Each row is one empirical fact of the paper and the model object it maps to. The coefficient column reports the estimate on standardized agro-similarity from the table and column the Source column names, with the standard error in parentheses and stars as in those tables; rows F1 to F6 are the estimates of Table 4, and a row without a numeric entry states the sign and whether the estimate is statistically significant. F19 reports the three birthplace groups in order: the same province, other outer-island provinces, and Java and Bali. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S8.1 Agricultural Employment Across Elasticity Pairs

The comparison varies the location-choice and sector-choice elasticities while retaining the same productivity experiment. Migration produces a larger agricultural-share decline than trade at 8 of the 17 pairs shown in Figure S11.

Figure S11: The Agricultural-Share Split at Every Solved Elasticity Pair



Notes: The 17 completed calibrations satisfying $\theta_L < \theta_S$ are ordered by θ_S and then θ_L. The economy has 342 locations, including the 134 cities in the main empirical sample, additional cities outside the exposure window, and rural district remainders. The experiment raises the land-augmenting agricultural productivity parameter by 0.05 log units at rural settlement hosts. Each response is the slope from regressing the city's agricultural-employment-share change, in percentage points, on standardized exposure with the main empirical controls. Migration alone holds food prices, traded-output prices, and traded-output employment at their unshocked values. Trade alone holds population at its unshocked value. The interaction equals the full response less the two restricted responses. The line and diamonds show the full response. Local services clear in every calculation. The fixed-price migration comparison differs from autarky and from the high-trade-cost comparison in the main text. These conditional model comparisons do not report sampling uncertainty. Source: the calibrated spatial model and the main empirical city sample.

S9 Theories of Structural Transformation

Two classical accounts explain what higher agricultural productivity does to the rest of the economy, and they make opposite predictions. In the first, cheaper food releases both spending and labor to the modern sector, and the economy transforms. In the second, the same productivity gain strengthens the region's comparative advantage in farming and pulls labor toward the land. In both cases, the positive productivity shock is within the same economy where outcomes are measured. A city that buys its food from the villages that received the gain is a small economy whose trading partner became more productive at farming, a case Matsuyama (1992) does not state; under his open-economy condition, the buyer's manufacturing share rises on impact. This appendix sets out both, states the condition that selects between them, and shows what the setups behind them contain. It proceeds in four steps: the closed-economy prediction and the two assumptions it turns on, the open-economy reversal and why the original models treat it as a switch, what the classical setups leave out that this design requires, and the frameworks that added locations, together with the three rival explanations of the same shock that the theory names.

S9.1 The Closed-Economy Prediction

The prediction that a better harvest transforms a nearby city descends from the dual-economy literature and rests on two assumptions. The first is Engel's law: the income elasticity of food demand is below one, so a household that feeds itself more cheaply moves its marginal spending to everything else. The second is that the food market clears at home, so that the extra food has nowhere to go but into local consumption.

Lewis (1954) states the price consequence of a better harvest. His two sectors exchange food against capitalist goods at an internal terms of trade, and because food demand is inelastic, a rise in the subsistence sector's productivity reaches the capitalist sector as a fall in the price of food and a rise in profit. In his model the transfer of labor runs through capitalist accumulation at a subsistence wage. The food-problem tradition states the second. In Gollin et al. (2002) households buy a subsistence quantity of food before valuing anything else, so labor stays on farms until the economy's food requirement is met and low agricultural productivity postpones industrialization (Gollin et al. 2007); the constraint appears already in Schultz (1953).

Matsuyama (1992) states the mechanism at its sharpest and shows how little it needs, and the subsistence preferences he used, the Stone-Geary form the model in this paper also uses, became the standard non-homotheticity of the structural-transformation literature. Labor is the only factor, preferences include a subsistence food requirement, and the food market clears at home. Food consumption then equals food output, and that single condition is sufficient to pin the manufacturing labor share as an increasing function of agricultural productivity. A better harvest frees workers for the modern sector because the extra food has nowhere else to go. Glaeser (2014) extends the argument to urbanization in poorer countries.

The prediction is about the employment composition, and it is worth separating from the scale prediction because the two are often stated together. Nothing in Matsuyama's closed-economy mechanism requires the city to grow. In Glaeser's closed economy, urbanization increases with farm productivity (his Proposition 1): his urban sector produces traded manufacturing goods, so a better harvest raises the share of workers in the city. Cheaper food raises what a household has left after feeding itself and tilts its marginal spending toward the modern good, and it lowers the return to farming inside the city relative to modern work; both forces move the employment composition without anyone changing location. The classical account that does require the city to grow is the labor one, and it is a different claim.

S9.2 The Open-Economy Reversal

The same model with both goods trading at world prices predicts the opposite. Matsuyama (1992) shows that when the food constraint is lifted, comparative advantage allocates labor instead, and an economy that becomes better at farming specializes in farming: the productivity gain pulls workers into agriculture and slows the transition out of it. The empirical literature has largely worked inside that case. Bustos et al. (2016) model each Brazilian region as a small open economy in which international prices pin agriculture's marginal product, so the direction of the local labor response is set by the factor bias of the technical change.

The reversal is a switch in these models because openness is a property of the whole country: both regimes are stated for an economy with one location, so there is no internal geography over which openness could vary. Matsuyama (2008) restates the caveat in the same terms. That is the feature the design changes, and the following subsection takes it up.

S9.3 Limits of the Classical Setups

I compare the twelve models this paper is positioned against on three facts: whether the setup contains a food market, whether it contains trade of any other kind, and whether it contains locations at a distance from each other. Locations are absent from every classical setup and arrive only with the quantitative spatial frameworks of the last decade, and that absence is why the theory has to be reinterpreted at the regional level before it can be used here. Table S19 records the three facts for each model.

Their absence has a consequence that is easy to miss, because the models do use place names. Harris & Todaro (1970) call one sector rural and the other urban, but the agricultural good sells at a single economy-wide terms of trade and the two sectors sit at no distance from each other. Gollin et al. (2016) do the same. A model with no distance between its locations has one food market by construction, so its food constraint binds for the country as a unit, and it cannot ask whether a particular city can be fed by a particular hinterland. The classical prediction is therefore about national agricultural productivity and national industrialization, and applying it at the level of a city and the villages that feed it changes the model.

I apply that regional interpretation to Indonesia's outer islands and report the travel-time gradients of district food-price gaps in the main text (Section 4.1).

S9.4 Spatial Frameworks and Their Open Questions

Five recent frameworks contain everything the classical setups leave out, and this paper differs from them in the source of the identifying variation. Eckert & Peters (2025) have discrete locations, costly trade in food, costly migration, and nonhomothetic demand; their frictions are calibrated, their productivities inverted from the static equilibrium, and their question is the incidence of secular structural change. Nagy (2023) is closest in spirit, with farm goods shipping from hinterlands to the cities they feed and city formation endogenous. His cities hold both farm and non-farm population, and he does not report a within-city agricultural response. Tombe & Zhu (2019) is the closest quantitative precedent for separating goods-market from labor-market integration, run on calibrated frictions and aggregate productivity. Pellegrina & Sotelo (2025) formalize the first stage of this paper's own design, farmers moving with crop-specific knowledge, and follow it into specialization and trade, leaving the urban system outside the model. Donaldson (2018) and Fajgelbaum & Redding (2022) establish how far shipping costs shape where production locates.

Two further theories predict the same direction of reallocation for reasons unrelated to the food market. In the price-effect family, structural change follows from sectors growing at different rates when the goods substitute poorly (Ngai & Pissarides 2007; Acemoglu & Guerrieri 2008; Herrendorf et al. 2014); an agricultural productivity gain then moves labor out of agriculture with no reference to who eats the output or where it is sold. In the income-effect family, to which the mechanism here belongs, the reallocation runs through nonhomothetic demand (Kongsamut et al. 2001; Boppart 2014; Comin et al. 2021). Both predict a fall in the agricultural employment share of an exposed city. The openness comparison supports the food-market mechanism if the other mechanisms respond similarly in the two groups of cities after controls. Migration, amenities, and rural demand could also vary with trade access.

S9.5 Three Rival Explanations of the Same Shock

The theory also names three ways a rural productivity gain can reach a city, and the paper's claim is about which one operates.

Labor. A labor-saving productivity gain would release settlers to the cities (Cao & Birchenall 2013; Bustos et al. 2016); rainfall-driven productivity gains raise local demand and move rural Indian labor into local non-farm work (Emerick 2018). Food trade can raise urban labor demand and induce migration from origins outside the settlements.

Capital. Bustos et al. (2020) follow the agricultural surplus out of the region it was earned in, as bank deposits that fund lending elsewhere, so agricultural productivity growth finances industrialization at a distance at a speed set by financial integration. Theirs is the one predecessor that crossed the boundary of the region that received the shock, and it did so through neither goods nor people.

Factor bias. Bustos et al. (2016) show that the direction of the local labor response depends on whether the technical change saves labor or augments land. A land-augmenting gain pulls labor into agriculture in the place that receives it, at a fixed outside price; the food market never enters. That is the same village-side pattern the food-trade account predicts for a different reason, so the factor bias of the shock is a genuine alternative explanation of the village side.

S9.6 Requirements for an Empirical Test

These mechanisms imply different responses in employment, population, and food prices. Within cities, the food expenditure share falls as spending per person rises (Section 5.2). The city's composition moves out of agriculture and into the modern sector (Section 5.1). Modern employment grows by more than total employment under the fixed-incumbent-workforce accounting (Section E.3). Net employment growth does not measure gross migration when arrivals replace departing workers or coincide with changes in labor-force participation. The estimate for the city's rice production less consumption is negative and statistically insignificant (Section 5.2). The village agricultural employment share has a positive, statistically insignificant gradient with agro-similarity. I find no evidence in the measured settler stay-share that settlers leave differentially by productivity. The measure does not follow the original settlers individually or include their locally born children (Section 5.2 and Section 5.3). The 2011 non-agricultural wage, overall wage and real wage estimates are statistically insignificant (Section 5.2). And the modern-sector response is concentrated in cities weakly connected to outside food markets and damped in cities that can already import cheaply, while the agricultural employment share falls in both groups (Section 5.2). The openness comparison supports the food-market mechanism if the other mechanisms respond similarly in the two groups of cities after controls. Migration, amenities, and rural demand could also vary with trade access.

Table S19: What Each Setup Contains, and Where Its Food Constraint Binds

Model	Food market	Other trade	Locations	Where the food constraint binds, and what the setup leaves open
Panel A: The classical setups: no locations				
Lewis (1954)	•	•	○	The economy as one unit; the terms of trade between food and capitalist goods move once, nationally. Later cases trade only between whole countries
Harris & Todaro (1970)	•	○	○	The country. Rural and urban are place names with no distance and no trade cost, and one national relative price of food. The urban sector has no agriculture, so the transformation can only arrive as migrants
Matsuyama (1992)	•	•	○	The country when closed, nowhere when open. Openness is a national switch, and with one location there is no migration to compete with the goods channel
Gollin et al. (2002)	•	○	○	The country as one unit, which is frictionless internal trade by assumption. The subsistence requirement is the mechanism and cannot be localized
Gollin et al. (2016)	•	•	○	The country; resource export earnings are an exogenous endowment. Their urban sector has no agriculture, so the within-city response is zero by construction
Bustos et al. (2016)	•	•	○	Nowhere: each region is a small open economy and food sells at the world price whatever its neighbors produce. Distance does not appear, and factor bias sets the sign
Bustos et al. (2020)	○	•	○	Nowhere in the model; the geography lives in the bank-branch data. Names the capital channel that the goods and labor accounts have to be separated from
Panel B: The frameworks with locations				
Eckert & Peters (2025)	•	•	•	The location, with food shipped at iceberg costs. Frictions are calibrated and productivities inverted from the static equilibrium, and the focus is the incidence of secular change
Tombe & Zhu (2019)	○	•	•	The region, through costly goods trade and costly migration. The closest precedent for separating the two integration costs, run on calibrated frictions and aggregate productivity
Pellegrina & Sotelo (2025)	○	•	•	The region, for agricultural goods. Formalizes the knowledge transfer behind this paper's first stage; the outcomes are specialization and trade, with no urban side
Nagy (2023)	•	•	•	The city and the hinterland that ships to it, which is this paper's unit. His cities hold both farm and non-farm population, and he does not report a within-city agricultural response
Tombe (2015)	•	•	•	The country, with trade costs largest in agriculture. The international statement of the localization argument; the units are whole countries

Notes: • marks an object the setup contains and ○ one it does not. "Food market" is a market in which the price of food is determined; "other trade" is trade in any other good, including trade with the rest of the world; "locations" means places at a measured distance from each other, so that a shipping or moving cost can be defined. Place names without distance, as in the rural and urban sectors of Harris & Todaro (1970), are not locations in this sense. The setups in Panel A are transcribed from the papers' own texts; the two Tombe rows and Tombe & Zhu (2019) are stated at the level of their published abstracts. Ordering within each panel is chronological.

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